

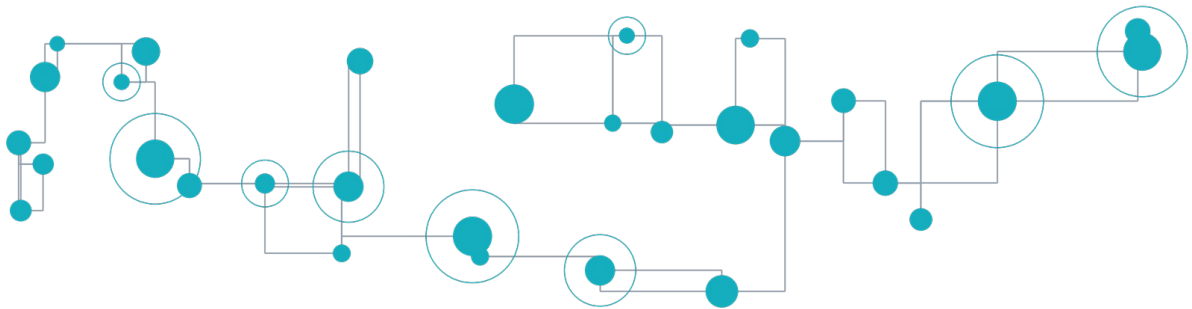


ROBOTICS & ARTIFICIAL INTELLIGENCE

CURRICULUM

A Research-Aligned, Age-Banded Curriculum for Ages 5–18

LEARN • INNOVATE • CREATE



CURRICULUM DEVELOPED BY Qasim Mushtaq, MS Robotics and AI

DOCUMENT STATUS For institutional review, accreditation submission, and instructor use

PAK-TURK MAARIF INTERNATIONAL SCHOOLS & COLLEGES

CHAK SHAHZAD CAMPUS • “Valuing Education”

“The role of the teacher is to create the conditions for invention rather than provide ready-made knowledge.”

SEYMOUR PAPERT

Foreword: How to Use This Document

This document is the official curriculum reference for the Pak-Turk Maarif Robotics & Artificial Intelligence Program. It substantially redevelops the school's original activity bank to meet the standard expected for institutional, regulatory, and accreditation review: every activity states a purpose, materials, procedure, assessable outcomes, the concepts it teaches, and its real-world relevance, and the whole program is mapped against recognised national and international frameworks (see Section 2).

The curriculum serves four age bands, Early Explorers (5–7), Junior Builders (8–10), Maker Engineers (11–15), and Innovators & Engineers (16–18), each running 30 hardware activities and 10 software activities, for 160 one-hour sessions in total. Every session produces a genuine artefact addressing a real-world problem rather than an abstract exercise, anchoring learning in application from a five-year-old's first session to an eighteen-year-old's capstone defence.

Default session length is 60 minutes, extending to 90 to 120 minutes for Capstone or paired-wireless activities at instructor discretion. Complexity scales by age, not by hardware availability, so the same procured inventory serves the entire 5–18 pipeline, and several real-world problems are deliberately revisited at increasing depth across bands (Section 4), consistent with Bruner's spiral curriculum model, in which core ideas are revisited with increasing sophistication as learners mature (Bruner, 1960).

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1. Program Philosophy & Pedagogical Foundations

1.1 Why Robotics and AI, and Why This Way

Robotics and AI sit at the intersection of engineering, computer science, and design. International evidence consistently finds that students learn these subjects most durably by building with them, not by reading about them. This curriculum rests on four converging research foundations.

Constructionism. Papert's constructionist theory holds that learners build robust understanding most effectively while constructing a tangible, shareable artefact, a claim first developed through his work on the LOGO programming language and later extended into educational robotics (Papert, 1980). Every activity therefore ends in a physical or digital artefact the student can demonstrate and keep, rather than a worksheet.

Computational thinking as a transferable literacy. Jeannette Wing's framing of computational thinking (decomposition, pattern recognition, abstraction, algorithm design) as a skill for everyone, not only computer scientists, underpins the program's insistence that a five-year-old sequencing picture-blocks and a sixteen-year-old tuning a PID control loop are practising the same cognitive moves, at an age-appropriate level of abstraction (Wing, 2006).

The engineering design cycle. From age eight onward, activities follow the same cycle used in the Next Generation Science Standards' engineering design strand: define the problem, develop and test solutions, and optimise based on evidence, teaching "trial and error" as a disciplined engineering practice rather than leaving it implicit (NGSS Lead States, 2013).

Project- and problem-based learning. A growing body of STEM-education research links well-scaffolded, real-world-anchored tasks to stronger retention, engagement, and skills transfer than lecture-based instruction (see, e.g., recent systematic reviews of PBL in STEM contexts). This is why every one of the 160 sessions builds something a student recognises from daily life, a doorbell, a streetlight, a weather app, rather than an abstract circuit exercise.

1.2 Guiding Principles

Application before abstraction. Every session opens with a concrete, real-world hook before any wiring or code is introduced, so students always know *why* they are building something before they learn *how*.

Scaffolded complexity, constant hardware. The same core inventory, Arduino Uno/Nano, ESP32, sensors, actuators, displays, and communication modules, is used from age 5 through 18. What changes is the level of abstraction: pre-wired picture-blocks at age 5–7, independently authored C/C++ and Python at age 16–18, not the equipment itself. This keeps procurement simple and lets older students mentor younger ones on shared equipment.

Spiral revisitation. Signature real-world problems (a night light, a security system, an irrigation controller) are deliberately revisited across the four age bands with rising technical depth, consistent with Bruner's argument that any subject can be taught honestly to a learner at any stage of development, provided it is revisited with increasing rigor (Bruner, 1960).

Assessable outcomes, not vague aims. Every activity states its outcomes as observable "Students will be able to..." statements, levelled against Bloom's Revised Taxonomy (Anderson & Krathwohl, 2001), so instructors assess directly against the activity rather than inferring what was meant.

Safety and progressive independence. Wiring responsibility, tool access (soldering irons, multimeters, relays), and unsupervised handling of flame/gas sensors are released gradually by age band (see each band's Safety Notes), matching adolescent development in fine-motor control and risk judgement.

Software and hardware as one pipeline, not two subjects. Part B (App/Web/Game Development) is not a separate "coding class" bolted onto robotics; most of its activities extend or control an artefact built in Part A, so students experience the full pipeline from physical sensor to on-screen dashboard, the real shape of IoT and robotics engineering work.

2. Alignment with International & National Frameworks

No single national curriculum yet fully codifies K–12 robotics and applied AI education, so this program is cross-referenced against the leading international computer-science, engineering-design, and AI-literacy frameworks, the national curriculum the school operates under, and the two most widely adopted robotics-competition frameworks. This gives accreditation panels, Ministry liaisons, and Maarif Foundation reviewers an independently verifiable evidence base, in a shared vocabulary with globally recognised standards.

Framework	Issuing Body	What It Contributes to This Program
ISTE Standards for Students: Computational Thinker, Innovative Designer, Creative Communicator	International Society for Technology in Education	Anchors the decompose, prototype, test, iterate, and communicate cycle used in every activity's "What Students Do" and "Real-World Relevance" fields (ISTE, 2024).
2026 CSTA PK–12 Computer Science Standards	Computer Science Teachers Association	Supplies the five concept strands this program's nine activity Domains are cross-walked against (CSTA, 2026).
AI Competency Framework for Students	UNESCO	Structures Age Group 3–4 AI & Data activities around its four competency dimensions and Understand → Apply → Create progression (UNESCO, 2024).
Next Generation Science Standards: Engineering Design (K-2-ETS1, 3-5-ETS1, MS-ETS1, HS-ETS1)	Achieve, Inc. / NGSS Lead States	Supplies the Define–Develop Solutions–Optimise design-cycle language used from Age Group 2 onward (NGSS Lead States, 2013).
National Curriculum in England: Computing Programmes of Study (Key Stages 1–4)	UK Department for Education	Benchmarks what "computational thinking" and "creating digital artefacts" should look like at each age, used to sanity-check this program's age bands (DfE, 2013).
National Curriculum of Pakistan: Computer Science, Grades 6–12 / Single National Curriculum	Government of Pakistan, National Curriculum Council & Federal Board (FBISE)	Ensures Age Group 3–4 outcomes report directly against the national subject curriculum the school is accountable to (National Curriculum Council, 2022).
World Robot Olympiad (WRO) age categories: WRO Starter, Elementary, Junior, Senior	WRO Association	Validates this program's age bands and offers a competitive pathway for students completing the Robotics & Actuation strand (WRO Association, 2026).
FIRST LEGO League: Discover, Explore, Challenge divisions	FIRST	A second age-banding reference (ages 4–16) and showcase pathway, particularly relevant to Age Groups 1–3 (FIRST, 2026).

Framework	Issuing Body	What It Contributes to This Program
Bloom's Revised Taxonomy	Anderson & Krathwohl (2001), after Bloom (1956)	Supplies the verb ladder (Remember → Understand → Apply → Analyse → Evaluate → Create) calibrating every activity's Learning Outcomes so cognitive demand rises within and across age bands.

How to read the crosswalk: each of the program's nine activity Domains corresponds to one or more CSTA and ISTE concept strands, and every AI & Data activity in Age Groups 3–4 sits at a specific UNESCO AI-competency dimension and progression level. Section 3 states the resulting Program-Level Learning Outcomes; Sections 5–8 show the mapping in individual activities.

3. Program-Level Learning Outcomes

On completing the full 5–18 pathway, a student who has progressed through the program will be able to:

1. **Design and build working physical-computing systems:** independently plan, wire, and program a multi-sensor, multi-actuator device that solves a self-identified real-world problem, and justify each design decision with evidence from testing. (*ISTE Innovative Designer; CSTA Algorithms & Programming; NGSS Engineering Design*)
2. **Reason computationally:** decompose an unfamiliar problem, recognise patterns from prior projects, abstract a general solution, and express it as an unambiguous, testable algorithm, whether in picture-blocks or C/C++/Python. (*Wing, 2006; ISTE Computational Thinker*)
3. **Work confidently across the physical-digital boundary:** read sensor data into software, and send software commands back out to actuators, understanding a device as one continuous system rather than separate “hardware” and “coding” tasks.
4. **Apply the engineering design cycle:** define a problem, prototype multiple candidate solutions, test them against evidence, and optimise iteratively, tolerating and learning from early-attempt failure. (*NGSS Engineering Design*)
5. **Communicate and defend technical work:** explain what a system does, why it was designed that way, and what its limitations are, to both technical and non-technical audiences, culminating in the Age Group 4 capstone defence. (*ISTE Creative Communicator*)
6. **Use data and, from Age Group 3 onward, AI responsibly:** collect, clean, visualise, and draw evidence-based conclusions from real sensor data, and by Age Group 4, build and critically evaluate a basic predictive model, while articulating the ethical and privacy implications of AI and biometric systems. (*UNESCO AI Competency Framework, dimensions 2–4*)
7. **Practise safe, professional engineering habits:** follow age-appropriate electrical, tool, and battery-handling safety procedures without prompting by the end of each band, building toward independent multimeter-verified practice by Age Group 4.
8. **Build and control networked systems:** from Age Group 3 onward, configure wireless communication (Bluetooth, NRF24L01, Wi-Fi/ESP32) between devices and, from Age Group 4, between a device and a cloud or web dashboard, and explain client-server communication in basic terms. (*CSTA Networks & the Internet*)
9. **Develop full-pipeline software products:** design, build, and test an application, webpage, or game that either stands alone or directly controls/visualises a hardware artefact built earlier in the same term.
10. **Transfer learning to a portfolio-ready or competition-ready outcome:** by age 16–18, scope, build, document, and present an original capstone project suitable for a university application portfolio, a national science fair, or entry to WRO/FIRST-LEGO-League-style competition.

4. Curriculum Architecture, Hardware Alignment & Progression Model

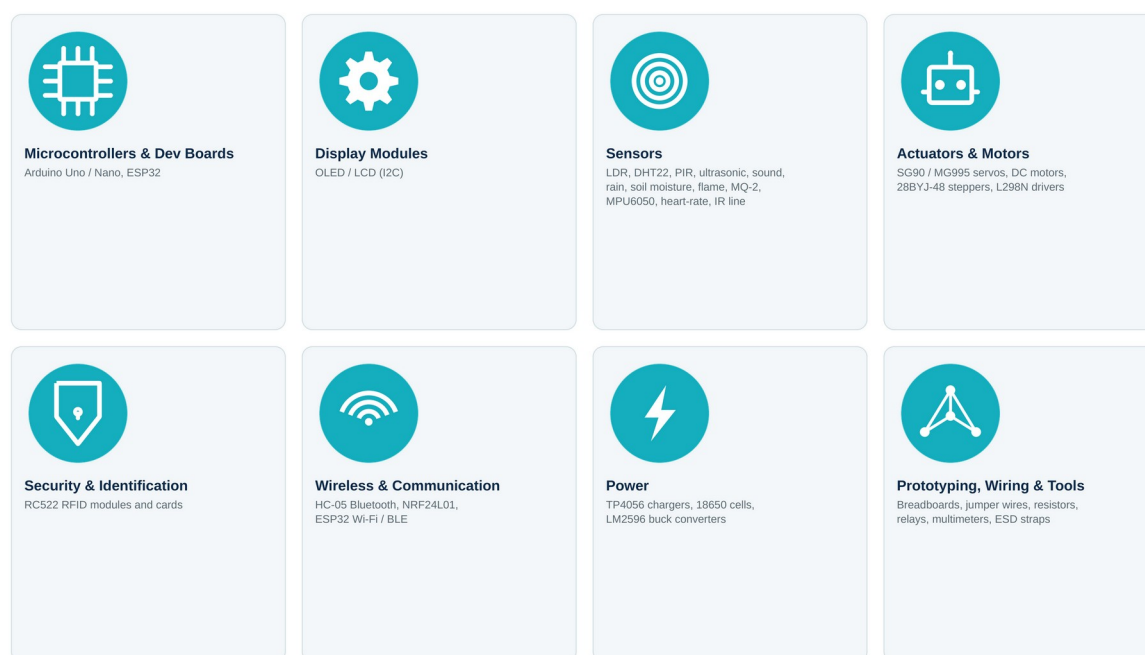
4.1 Age Bands at a Glance

Age Band	Name	Sessions	Primary Coding Tools	Core Shift
5–7	Early Explorers	40 (30 hardware + 10 software)	Tinkercad Circuits, ArduBlock/mBlock 5, ScratchJr/Scratch	Cause-and-effect, guided sensory play, no code syntax
8–10	Junior Builders	40 (30 hardware + 10 software)	mBlock5 (block-to-code bridge), Arduino IDE (starter sketches), Scratch, MIT App Inventor	First real breadboard wiring; block-to-text coding bridge
11–15	Maker Engineers	40 (30 hardware + 10 software)	Arduino IDE (C/C++), Thonny (MicroPython/ESP32), MIT App Inventor, Python, HTML/CSS/JS	Independent C/C++; first wireless/IoT work
16–18	Innovators & Engineers	40 (30 hardware + 10 software)	Arduino IDE, MicroPython/Thonny, Python (pandas/matplotlib/scikit-learn), Node.js/Flask, HTML/CSS/JS, App Inventor (advanced)/Flutter, Unity/Godot/Pygame	Control theory (PID), ML, full-stack, open capstone

Total: **160 unique, one-hour, objective-driven activities** across the full 5–18 pathway, enough for a full academic year per band at 1–2 sessions/week without repeating an activity (see Section 10).

4.2 Hardware Inventory Alignment

The program is deliberately built around one shared inventory rather than age-specific kits, so procurement, storage, and lab-technician workflows stay simple across all four bands:



- **Microcontrollers & Dev Boards:** Arduino Uno/Nano, ESP32
- **Display Modules:** OLED/LCD (I2C)
- **Sensors:** LDR, DHT22 (temperature/humidity), PIR (motion), ultrasonic (HC-SR04), sound (KY-037), rain/water-drop, soil moisture, flame, MQ-2 (gas), MPU6050 (gyroscope/accelerometer), heart-rate pulse, IR line sensors
- **Actuators & Motors:** SG90/MG995 servos, DC motors, 28BYJ-48 steppers with ULN2003 drivers, L298N motor drivers
- **Security & Identification:** RC522 RFID modules and cards
- **Wireless & Communication:** HC-05 Bluetooth, NRF24L01, ESP32 Wi-Fi/BLE
- **Power:** TP4056 chargers, 18650 cells, LM2596 buck converters
- **Prototyping, Wiring & Tools:** breadboards, jumper wires, resistors, relays, multimeters, ESD wrist straps

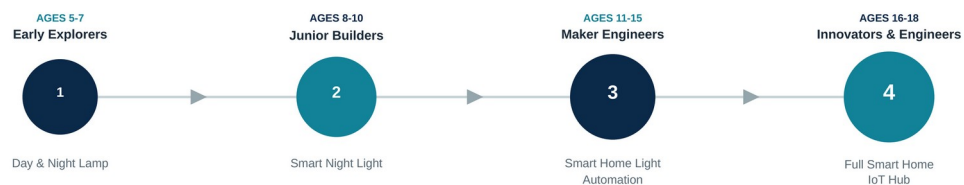
Every activity's "Materials Required" field draws on exactly this list, so lab technicians can pull a kit per session without reinterpreting the activity, and the same relay module a seven-year-old watches a teacher operate becomes, by age sixteen, a component the student wires and code-drives independently.

4.3 Spiral Progression Threads

The following real-world problem families are intentionally revisited at rising depth across the four bands, giving returning students a familiar anchor while raising the technical bar each time (Bruner, 1960):

Real-World Thread	Age 5–7	Age 8–10	Age 11–15	Age 16–18
Automatic lighting	Day & Night Lamp	Smart Night Light	Smart Home Light Automation	Full Smart Home IoT Hub

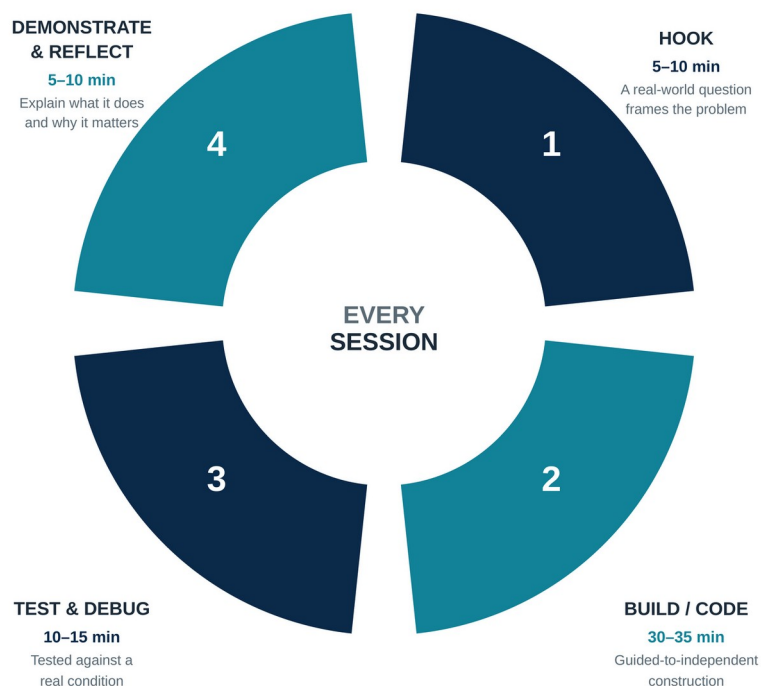
Real-World Thread	Age 5–7	Age 8–10	Age 11–15	Age 16–18
Environmental monitoring	Hot or Cold Helper	Mini Weather Station Display	Multi-Sensor Weather Dashboard	Sensor Data Logging & Visualisation Pipeline
Agriculture / irrigation	Thirsty Plant Alarm	Plant Watering Reminder	Smart Irrigation System	Predictive Plant Health Analytics
Security & access	Scaredy-Cat Alarm	Pet Door Motion Alarm	RFID Access Control System	Multi-Factor Security System / Smart Locker Access
Mobility & obstacle-avoidance	Treasure Hunt Beeper	Obstacle-Avoiding Toy Car	Obstacle-Avoiding Autonomous Robot	Autonomous Obstacle-Avoidance Robot with Speed Control
Health monitoring	Feel My Heartbeat	Heart Rate Display Gadget	Heart Rate Monitor with Wireless Alert	BLE Health Wearable Prototype
Structural/ vibration safety	Shake-a-Maraca	Earthquake Shake Alarm	Structural Vibration Health Monitor	Structural Health Monitoring with Cloud Alerts
Dashboards & connected software	Traffic Light Simulator	Weather Dashboard Webpage	ESP32 Sensor Dashboard Web App	Full-Stack IoT Dashboard



REAL-WORLD THREAD: AUTOMATIC LIGHTING · REVISITED AT RISING DEPTH ACROSS ALL FOUR AGE BANDS

4.4 The 60-Minute Session Model

Every activity, regardless of age band, follows the same four-phase structure, deliberately mirroring the engage–build–test–reflect rhythm of the engineering design cycle so that the *shape* of a lesson is predictable even as its *content* matures:

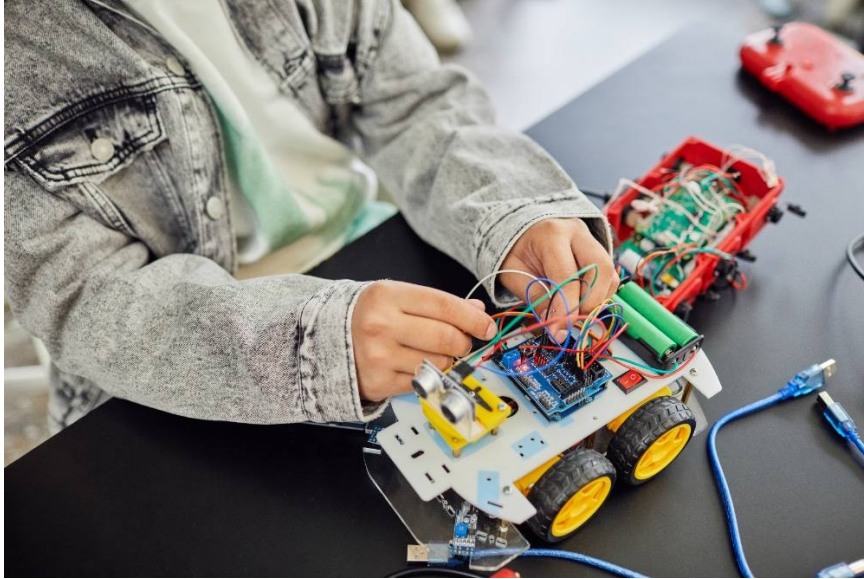


1. **Hook (5–10 min):** a real-world question or story frames the problem before any hardware is touched.
2. **Build/Code (30–35 min):** guided-to-independent (by age) construction of the circuit, mechanism, or program.
3. **Test & Debug (10–15 min):** the artefact is tested against a real condition; younger bands test with adult support, older bands run a written test plan.
4. **Demonstrate & Reflect (5–10 min):** the student demonstrates the finished build and explains, in their own words (or in a short written log from Age Group 2 onward), what it does and why it matters, the direct evidence base for the assessment rubric in Section 9.

Capstone and multi-board wireless-pair activities extend this into a 90–120 minute double session at instructor discretion, as in the original program design.

AGE BAND 01

EARLY EXPLORERS

Ages 5–7

Guided sensory play and cause-and-effect discovery, told entirely through picture-block programming.



30 HARDWARE-BASED ACTIVITIES • 10 APP / WEB / GAME ACTIVITIES

5. Age Group 1 (Ages 5-7): Early Explorers

At this age, children learn best through guided sensory play, repetition, and cause-and-effect discovery rather than text-based code. All circuits in this band are pre-wired or assembled with heavy facilitator support; children snap, plug, and press rather than solder or wire from scratch. Programming is done entirely through drag-and-drop picture blocks (ArduBlock or mBlock5, or the Tinkercad Circuits simulator before any physical build), so no reading of code syntax is required. Every session opens with a story or real-world hook and closes with the child explaining, in their own words, what their invention does and why it matters, the direct evidence of learning used in the Section 9 rubric.

Primary Coding Tools: Tinkercad Circuits (simulation), ArduBlock / mBlock5 (picture-block programming for Arduino Uno), ScratchJr / Scratch (screen-based apps)

Safety Notes: Facilitator pre-wires or supervises 100% of wiring. No soldering. Batteries limited to 9V/coin-cell with holders. All flame/gas sensor demonstrations are teacher-operated only.

Part A: 30 Hardware-Based Activities (Ages 5-7)

Each session is 60 minutes and ends with the student demonstrating the finished build and explaining its real-world purpose in their own words.



1. Traffic Light Helper: Electronics & Circuits

Materials Required: 3x LED (red/yellow/green), resistors, breadboard, Arduino Uno (ArduBlock)

Purpose & Objective: Students build and sequence a working three-colour traffic light, learning that lights can be told what order to switch on in.

What Students Do: After a story about a busy road needing a helper, students plug LEDs into a pre-wired breadboard section with adult guidance, then snap together ArduBlock picture-blocks to switch the red, yellow, and green lights on in the correct order and demonstrate the cycle to the class.

Learning Outcomes (Students will be able to):

- Identify the three traffic-light colours and explain what each one means for cars and walkers.
- Connect a simple LED circuit on a breadboard with adult guidance.
- Sequence picture-blocks so the lights turn on in the correct order.

Concepts & Skills Covered: circuits, LEDs, colour meaning, sequencing

Real-World Relevance: Real traffic lights use this same on/off sequencing logic to keep roads safe every day.



2. Ring My Doorbell: Electronics & Circuits

Materials Required: push button, active buzzer, breadboard, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a working push-button doorbell to discover that pressing a button can complete, or “close,” an electrical circuit.

What Students Do: Students discuss what happens when a visitor arrives at their own front door, then assemble a button-to-buzzer circuit with facilitator support and take turns “answering the door” by pressing the button to sound the buzzer.

Learning Outcomes (Students will be able to):

- Explain that pressing the button completes (closes) the circuit.
- Build a simple button-to-buzzer circuit with support.
- Describe what a real doorbell is used for at home.

Concepts & Skills Covered: push buttons, open/closed circuits, cause and effect

Real-World Relevance: Every home doorbell, simple or smart, still relies on this same button-closes-circuit idea.



3. Light-Up Greeting Card: Electronics & Circuits

Materials Required: assorted LEDs, coin/9V battery with holder, copper tape or switch, paper craft supplies

Purpose & Objective: Students build a simple paper circuit so an LED lights up when a handmade card is opened, learning to identify battery polarity.

What Students Do: Students decorate a paper card, then attach an LED and battery using copper tape or a simple switch with adult guidance, testing which way the battery must face for the light to turn on before giving the finished card to a family member.

Learning Outcomes (Students will be able to):

- Identify the positive and negative sides of a battery.
- Assemble a simple paper circuit that lights up an LED.
- Create a personalised card and give it to a real family member.

Concepts & Skills Covered: battery polarity, paper circuits, series circuits

Real-World Relevance: Greeting cards, packaging, and some wearable products use this same low-cost paper-circuit technique.



4. Clap to Light Up: Sensors & Environment

Materials Required: sound sensor (KY-037), LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students trigger an LED with a hand clap, discovering that sensors let a device “hear” sound and react to it.

What Students Do: Students test the sound sensor by clapping at different volumes and watching the LED respond, then work in pairs to compare whose clap turns the light on fastest.

Learning Outcomes (Students will be able to):

- Describe how a sound can turn a light on.
 - Test the sensor by clapping at different volumes.
 - Connect the idea of cause (clap) and effect (light).
-

Concepts & Skills Covered: sound sensors, analog input, cause and effect

Real-World Relevance: Clap-activated lights are a real consumer product found in bedrooms and hallways worldwide.



5. Day & Night Lamp: Sensors & Environment

Materials Required: LDR light sensor, LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a lamp that switches itself on in the dark, learning why streetlights don't need a person to turn them on.

What Students Do: Students shine a torch on the light sensor and then cover it with a hand, watching the LED turn on automatically as the room gets "darker," then explain in their own words why the lamp knows when to switch on.

Learning Outcomes (Students will be able to):

- Explain why a streetlamp turns on when it gets dark.
- Observe how covering the sensor changes its reading.
- Build a lamp that switches on by itself in the dark.

Concepts & Skills Covered: light sensors (LDR), automatic switching, day/night

Real-World Relevance: Dusk-to-dawn streetlights use this exact sensing idea to save electricity across entire cities.



6. Rainbow Colour Mixer: Electronics & Circuits

Materials Required: assorted LEDs (red, green, blue, yellow, white), push buttons, breadboard

Purpose & Objective: Students press buttons to light up different coloured LEDs, building early vocabulary for colour and circuits.

What Students Do: Students take turns pressing each button to see which LED lights up, then work together to light up two colours at once and describe what they notice about how the light looks.

Learning Outcomes (Students will be able to):

- Name several colours of light.
- Press buttons to light up different coloured LEDs together.
- Describe what they notice when different colours sit side by side.

Concepts & Skills Covered: colour vocabulary, buttons, parallel circuits

Real-World Relevance: Coloured LED indicator lights are used on toys, appliances, and devices to signal different states.



7. Waving Robot Friend: Robotics & Actuation

Materials Required: SG90 micro servo, craft arm, push button, Arduino Uno (ArduBlock)

Purpose & Objective: Students attach a paper arm to a servo motor and make it wave, learning what a motor does in simple words.

What Students Do: Students decorate a craft robot arm, attach it to the servo, and press a button to trigger a wave, then greet a partner with their finished “robot friend.”

Learning Outcomes (Students will be able to):

- Describe in simple words what a servo motor does.
- Press a button to make a craft arm wave.
- Relate the activity to robots that greet people.

Concepts & Skills Covered: servo motors, rotation, buttons as triggers

Real-World Relevance: Waving, greeting robots are already used as real receptionists in airports, hotels, and hospitals.



8. Hot or Cold Helper: Sensors & Environment

Materials Required: DHT22 sensor, red/blue LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a device that lights red for warm and blue for cold, learning to read a simple temperature indicator.

What Students Do: Students warm the sensor with their hands and watch the reading and LED colour change, then test it again after letting it cool and record which colour appeared each time.

Learning Outcomes (Students will be able to):

- Tell the difference between something hot and something cold.
- Watch a sensor reading change when warmed by a hand.
- Choose the correct LED colour for the temperature shown.

Concepts & Skills Covered: temperature sensors, hot/cold vocabulary, thresholds

Real-World Relevance: This same hot/cold sensing idea is used in thermostats and fever thermometers at home.



9. Feed the Teddy: Robotics & Actuation

Materials Required: SG90 servo, small cup/spoon attachment, push button, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a pretend feeding machine that dispenses a treat for a toy when a button is pressed.

What Students Do: Students press the button to watch the servo tip the spoon and “feed” a teddy bear, then discuss how a real pet at home gets fed and by whom.

Learning Outcomes (Students will be able to):

- Explain that pressing a button can make a motor move.
- Watch a servo “dispense” a pretend treat for a toy.

- Relate the project to feeding a real pet at home.

Concepts & Skills Covered: servo motors, dispensing mechanisms, button triggers

Real-World Relevance: Automatic pet feeders use this same button-or-timer-to-servo idea in real households.



10. Treasure Hunt Beeper: Sensors & Environment

Materials Required: ultrasonic sensor (HC-SR04), buzzer, Arduino Uno (ArduBlock)

Purpose & Objective: Students use a distance-sensing “treasure detector” to find a hidden object, learning that sensors can measure how far away something is.

What Students Do: A teacher hides a “treasure” in the room; students walk around holding the sensor and listen for the beeping to speed up as they get closer, comparing strategies with a partner.

Learning Outcomes (Students will be able to):

- Describe that the sensor can tell how far away something is.
- Notice the beeping gets faster when closer to an object.
- Use the device to find a hidden “treasure” in the room.

Concepts & Skills Covered: ultrasonic distance sensing, feedback, proximity

Real-World Relevance: The same ultrasonic sensing helps real cars detect how close they are to a wall while parking.



11. Scaredy-Cat Alarm: Sensors & Environment

Materials Required: PIR motion sensor, buzzer, LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a simple motion alarm and test it by walking past, learning that sensors can notice movement nearby.

What Students Do: Students take turns walking past the sensor at different distances and speeds, observing when the alarm triggers, then discuss where in a real home this kind of alarm would be useful.

Learning Outcomes (Students will be able to):

- Explain that the sensor notices when something moves nearby.
- Test the alarm by walking past it.
- Relate the project to keeping a home safe.

Concepts & Skills Covered: motion sensors (PIR), alarms, safety

Real-World Relevance: PIR motion sensors are the core of most household security lights and burglar alarms.



12. Umbrella Helper: Sensors & Environment

Materials Required: rain/water-drop sensor, LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a device that lights up when it detects water, helping decide whether an umbrella is needed.

What Students Do: Students drip a few drops of water onto the sensor and watch the LED respond, then role-play “checking the weather” before pretending to leave the house.

Learning Outcomes (Students will be able to):

- Describe how the sensor detects water drops.
- Decide whether an umbrella is needed by reading the LED.
- Connect the activity to checking the weather before going out.

Concepts & Skills Covered: rain sensors, moisture detection, decision-making

Real-World Relevance: Automatic car wipers and smart irrigation systems use the same water-detection principle.



13. Thirsty Plant Alarm: Sensors & Environment

Materials Required: soil moisture sensor, LED, buzzer, Arduino Uno (ArduBlock)

Purpose & Objective: Students build an alarm that tells them when a classroom plant needs water, learning why plants need water to stay healthy.

What Students Do: Students test the sensor in dry soil and then in watered soil, comparing the alarm’s response, and use it over several days as a real reminder to water the class plant.

Learning Outcomes (Students will be able to):

- Explain why plants need water to stay healthy.
- Observe how wet vs. dry soil changes the sensor reading.
- Care for a real classroom plant using the alarm as a reminder.

Concepts & Skills Covered: soil moisture sensors, plant care, thresholds

Real-World Relevance: Commercial smart-irrigation systems use exactly this sensor to decide when farmland needs watering.



14. Fire Truck Siren: Electronics & Circuits

Materials Required: active buzzer, push button, breadboard

Purpose & Objective: Students build a simple button-activated siren and connect it to the idea of a firefighter’s job.

What Students Do: Students build the button-to-buzzer circuit with growing independence from the teacher, then take turns operating the “siren” safely and politely while discussing what firefighters do to help people.

Learning Outcomes (Students will be able to):

- Build a simple button-to-buzzer circuit with growing independence.
- Describe what a firefighter's job is.
- Take turns operating the siren safely and politely.

Concepts & Skills Covered: buzzers, buttons, community helpers

Real-World Relevance: Real emergency vehicles use driver-triggered sirens to warn others and clear a path quickly.



15. Counting Blinks Game: Programming Logic

Materials Required: LED, push button, Arduino Uno (ArduBlock)

Purpose & Objective: Students play a counting game where each button press makes an LED blink once, connecting number sense to a simple programmed rule.

What Students Do: Working with a partner, students take turns pressing the button and counting the blinks aloud, then sequence a simple “if pressed, blink” picture-block rule themselves.

Learning Outcomes (Students will be able to):

- Count button presses out loud with a partner.
- Sequence a simple “if pressed, blink” picture-block rule.
- Practise number recognition through hands-on play.

Concepts & Skills Covered: conditional logic (if), counting, sequencing

Real-World Relevance: This is a child-friendly first taste of the if-then logic that runs every app and website.



16. Railway Crossing Gate: Robotics & Actuation

Materials Required: SG90 servo (gate arm), ultrasonic sensor, buzzer, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a model railway gate that closes automatically when a “train” approaches, learning why safety barriers matter.

What Students Do: Students pass an object near the ultrasonic sensor and watch the servo-driven gate arm swing down with a warning buzz, then discuss why real trains need this kind of automatic safety gate.

Learning Outcomes (Students will be able to):

- Describe why a train crossing needs a safety gate.
- Trigger the gate to close using the distance sensor.
- Explain why safety barriers matter near roads and railways.

Concepts & Skills Covered: servo motors, distance sensing, road/rail safety

Real-World Relevance: Real level-crossing gates use the same sense-then-actuate logic to protect drivers and pedestrians.



17. Shake-a-Maraca: Sensors & Environment

Materials Required: MPU6050 gyroscope/accelerometer, buzzer, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a shake-activated musical instrument, discovering that motion itself can be sensed and turned into sound.

What Students Do: Students shake a decorated craft shaker fitted with the sensor and listen for the buzzer sound, then perform a simple rhythm pattern for the class.

Learning Outcomes (Students will be able to):

- Describe how shaking an object creates motion.
- Connect a shaking action to a sound effect.
- Create and play a simple musical instrument.

Concepts & Skills Covered: motion sensors, accelerometers, cause and effect

Real-World Relevance: Motion-triggered sound is the same idea behind fitness trackers and motion-controlled game controllers.



18. Feel My Heartbeat: Sensors & Environment

Materials Required: heart rate pulse sensor, LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students find their own pulse and compare it to an LED blinking in time with a heart-rate sensor.

What Students Do: Students find their pulse with a finger on their wrist, then place a finger on the pulse sensor and watch the LED blink with each heartbeat, discussing what the heart does in simple words.

Learning Outcomes (Students will be able to):

- Find their own pulse using a finger on their wrist.
- Watch an LED blink in time with their heartbeat.
- Describe, in simple words, what the heart does.

Concepts & Skills Covered: pulse sensors, the human heart, cause and effect

Real-World Relevance: This is the same basic sensing idea used in fitness watches and hospital heart monitors.



19. Light-Up Story Jar: Electronics & Circuits

Materials Required: LED, battery holder, switch, decorative jar craft

Purpose & Objective: Students build a switch-controlled light-up jar and use it as a prop to tell a short story.

What Students Do: Students build a simple switch circuit independently, decorate a jar around it, and then tell a short story to the class, switching the jar's light on at a key moment.

Learning Outcomes (Students will be able to):

- Build a simple switch circuit independently.
- Decorate a “story jar” that lights up when switched on.
- Tell a short story to the class using the jar as a prop.

Concepts & Skills Covered: switches, series circuits, storytelling

Real-World Relevance: The same switch logic controls every lamp, torch, and toy that has an on/off button.



20. Animal Sound Box: Electronics & Circuits

Materials Required: 4x push buttons, buzzer, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a multi-button sound box where each button plays a different sound, then perform an “animal show.”

What Students Do: Students match each button to an animal sound pattern with facilitator support, rehearse pressing them in order, then perform a short animal show for classmates.

Learning Outcomes (Students will be able to):

- Match each button to a different sound pattern.
- Build a simple multi-button circuit with support.
- Create and perform a short “animal show” for classmates.

Concepts & Skills Covered: multiple inputs, buzzers, performance/sequencing

Real-World Relevance: Sound buttons like these are the same technology found in children’s toys and greeting cards.



21. Quick Hands Reaction Game: Electronics & Circuits

Materials Required: push button, LED, Arduino Uno (ArduBlock)

Purpose & Objective: Students play a reaction-time game, pressing a button as fast as they can when a light comes on.

What Students Do: Students take turns reacting to the LED cue as quickly as possible while a partner counts and compares scores fairly, then discuss what “reaction time” means.

Learning Outcomes (Students will be able to):

- React to a light cue by pressing a button quickly.
- Take turns and count scores fairly with a partner.
- Describe what “reaction time” means in their own words.

Concepts & Skills Covered: stimulus-response, fairness/turn-taking, timing

Real-World Relevance: Reaction-time games like this are a simplified version of tests used in real sports-science and driving research.



22. Spinning Pinwheel: Mechanical Design

Materials Required: DC motor (3-6V), battery holder, switch, craft pinwheel blades

Purpose & Objective: Students build a simple motor-powered pinwheel and observe how electricity makes it spin.

What Students Do: Students build a motor circuit with a switch, attach craft pinwheel blades, and flip the switch to watch it spin, comparing it to real fans and windmills.

Learning Outcomes (Students will be able to):

- Build a simple motor circuit with a switch.
- Observe how electricity makes the motor spin.
- Relate the spinning motion to real fans and windmills.

Concepts & Skills Covered: DC motors, switches, rotational motion

Real-World Relevance: The same spinning-motor principle powers household fans and, at larger scale, wind turbines.



23. Magic Door for Dolls: Robotics & Actuation

Materials Required: SG90 servo, push button, small craft door, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a working automatic door for a dollhouse, connecting a button press to an opening mechanism.

What Students Do: Students attach a craft door to the servo arm, press the button to watch it swing open, and discuss how automatic doors work in real shops.

Learning Outcomes (Students will be able to):

- Connect a button press to an opening door.
- Describe how automatic doors work in real shops.
- Build a working automatic door for a dollhouse.

Concepts & Skills Covered: servo motors, actuation, everyday automation

Real-World Relevance: Automatic sliding doors at shops and hospitals use sensors and motors in exactly this way.



24. Flashlight Colour Hunt: Sensors & Environment

Materials Required: LDR sensor, LED, flashlight

Purpose & Objective: Students use a light sensor to sort classroom objects into “light” and “dark,” learning how a sensor “senses” brightness.

What Students Do: Students shine a flashlight on the sensor and observe the change, then test several classroom objects and surfaces, sorting them by how much light they reflect.

Learning Outcomes (Students will be able to):

- Shine a light on the sensor and observe the change.
- Sort classroom objects into “light” and “dark” using the sensor.
- Describe how the sensor “senses” brightness.

Concepts & Skills Covered: light sensors, classification, observation

Real-World Relevance: Light sensors like this automatically dim or brighten phone and laptop screens.



25. Light Show Band: Electronics & Circuits

Materials Required: assorted LEDs, buzzer, push buttons, breadboard

Purpose & Objective: Students sequence buttons to create their own simple light-and-sound show.

What Students Do: Students plan a short sequence of button presses to create a light-and-sound pattern, practise it, and describe the order of their show to a partner.

Learning Outcomes (Students will be able to):

- Sequence buttons to create a simple light-and-sound show.
- Describe the order of their show to a partner.
- Practise fine-motor skills connecting parts with adult help.

Concepts & Skills Covered: sequencing, multiple outputs, fine-motor circuit building

Real-World Relevance: Sequenced lights and sounds are the same basic idea behind stage lighting and simple toy shows.



26. My Weather Picture Book: Sensors & Environment

Materials Required: DHT22 sensor, 2x LED (hot/cold), Arduino Uno (ArduBlock)

Purpose & Objective: Students read a simple temperature indicator and use it to make a one-page weather picture book.

What Students Do: Students read the hot/cold LED indicator, match weather words to LED colours, and draw and label a one-page picture book about the day's weather.

Learning Outcomes (Students will be able to):

- Read a simple temperature indicator light.
- Match weather words (hot, cold, warm) to LED colours.
- Create a one-page weather picture book.

Concepts & Skills Covered: temperature sensors, vocabulary, data-to-picture translation

Real-World Relevance: This mirrors how real weather apps turn sensor data into simple icons anyone can read.



27. Robot Friend Face: Electronics & Circuits

Materials Required: 2x LED (eyes), push button, craft robot face, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a robot face with two LED “eyes” that wake up when a button is pressed, then give it a personality.

What Students Do: Students build the two-LED circuit, attach it behind a craft robot face, press the button to “wake” the robot, and describe their robot’s personality in one sentence.

Learning Outcomes (Students will be able to):

- Build a circuit with two LEDs that act as “eyes.”
- Press a button to “wake up” the robot face.
- Describe their robot’s personality in one sentence.

Concepts & Skills Covered: parallel LED circuits, buttons, creative design

Real-World Relevance: Expressive robot “faces” like this are used in real social and educational robots for children.



28. Shake-to-Ring Door Bell: Sensors & Environment

Materials Required: MPU6050 sensor, buzzer, Arduino Uno (ArduBlock)

Purpose & Objective: Students compare a shake-triggered bell to a button-triggered bell, exploring two different ways to trigger the same sound.

What Students Do: Students test the sensor with gentle shakes and note how the sound triggers, then compare it side-by-side to a button bell from an earlier lesson and discuss the difference.

Learning Outcomes (Students will be able to):

- Describe the difference between a button bell and a shake bell.
- Test the sensor with gentle shakes.
- Compare two different ways to trigger the same sound.

Concepts & Skills Covered: motion sensors, comparison, alternative triggers

Real-World Relevance: Many real devices, like phones, let users choose between a tap, a shake, or a button to trigger an action.



29. Waving Greeting Card Robot: Robotics & Actuation

Materials Required: SG90 servo, push button, pop-up card craft, Arduino Uno (ArduBlock)

Purpose & Objective: Students build a pop-up card with a servo-powered waving arm as a gift for someone real.

What Students Do: Students attach a paper arm to the servo inside a pop-up card craft, press the button to trigger a wave for a classmate, then give the finished card to a real person as a gift.

Learning Outcomes (Students will be able to):

- Attach a paper arm to a servo motor.
- Trigger a wave for a classmate using a button.
- Give the finished card to a real person as a gift.

Concepts & Skills Covered: servo motors, mechanical linkages, giving/sharing

Real-World Relevance: This combines the same servo-arm mechanism used in simple animatronics and greeting gadgets.



30. Circuit Treasure Hunt Stations: Electronics & Circuits

Materials Required: breadboards, LEDs, switches, batteries (multiple stations)

Purpose & Objective: Students revisit several earlier circuit ideas at once by completing a simple circuit at each of several classroom stations.

What Students Do: In small groups, students rotate through multiple pre-set stations, completing a simple circuit at each to light it up, recalling parts and vocabulary from earlier lessons as they go.

Learning Outcomes (Students will be able to):

- Complete a simple circuit at each station to light it up.
- Recall the parts used in earlier lessons by name.
- Work together as a class to light up every station.

Concepts & Skills Covered: circuit review, vocabulary recall, teamwork

Real-World Relevance: This capstone-style review consolidates a full term of circuit vocabulary before students move to more independent building.

Part B: 10 App / Web / Game Development Activities (Ages 5-7)

These sessions build foundational software-creation skills in parallel with the hardware track. Several explicitly extend or control an earlier hardware build, so students experience the complete hardware-to-software pipeline.



31. My First Animated Story: Programming Logic

Materials Required: ScratchJr (tablet or computer)

Purpose & Objective: Students sequence a few blocks to animate a character and tell a very short story with a beginning and an end.

What Students Do: Students choose a character, sequence 3-4 motion/sound blocks to animate it, then share their finished animation with the class.

Learning Outcomes (Students will be able to):

- Sequence 3-4 blocks to animate a character.
- Tell a simple story with a clear beginning and end.
- Share their finished animation with the class.

Concepts & Skills Covered: sequencing, story structure, ScratchJr blocks

Real-World Relevance: This is the earliest version of the sequencing logic used in every animated film and app.



32. Catch the Falling Fruit: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students build a simple catching game where a sprite moves left and right to catch falling objects.

What Students Do: Students use motion blocks to move their sprite, test their game by trying to catch falling fruit, then suggest one change to improve it.

Learning Outcomes (Students will be able to):

- Use motion blocks to move a sprite left and right.
- Recognise a simple “catch the falling object” game goal.
- Play and suggest one change to their own game.

Concepts & Skills Covered: motion blocks, game goals, iteration

Real-World Relevance: “Catching” games like this are the basic pattern behind many popular children’s mobile games.



33. Button Piano App: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students assign different sounds to different key-presses to build their own simple instrument app.

What Students Do: Students assign a sound to each key-press block, test their instrument, and perform a short tune for a partner.

Learning Outcomes (Students will be able to):

- Assign a different sound to each key-press block.
- Build a simple working instrument app.
- Perform a short tune for a partner.

Concepts & Skills Covered: event blocks, sound blocks, key-press input

Real-World Relevance: This is the same key-to-sound mapping used by real digital piano and music apps.



34. Animated Birthday Card: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students build a digital birthday card that animates and displays a personal typed message.

What Students Do: Students add costume changes to animate a character, type a personal birthday message into their card app, then share the finished card with a real person.

Learning Outcomes (Students will be able to):

- Add costume changes to animate a character.
- Type a personal message into their card app.
- Share their finished card with a real person.

Concepts & Skills Covered: costumes/animation frames, text input, personalisation

Real-World Relevance: Digital greeting cards and e-cards use exactly this animate-and-personalise pattern.



35. Traffic Light Simulator: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students build an on-screen traffic light simulator using a repeat loop, connecting it to their earlier hardware build.

What Students Do: Students use a repeat loop to cycle the light colours in order, connect the idea back to the earlier Traffic Light Helper hardware build, and explain why order matters.

Learning Outcomes (Students will be able to):

- Use a repeat loop to cycle colours in order.
- Connect the app to the earlier Traffic Light Helper hardware build.
- Explain why order matters in a sequence of steps.

Concepts & Skills Covered: repeat loops, sequencing, hardware-software connection

Real-World Relevance: This is students' first hands-on encounter with the loop structure that underlies nearly all software.



36. Weather Puppet Show: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students build a costume-changing weather puppet and narrate a simple weather report.

What Students Do: Students change a sprite's costume to show different weather types, then narrate a short weather report to the class relating it to the real weather outside.

Learning Outcomes (Students will be able to):

- Change a sprite's costume to show different weather.
- Narrate a simple weather report to the class.
- Relate the app to the real daily weather outside.

Concepts & Skills Covered: costume-changing, narration, real-world observation

Real-World Relevance: This mirrors how real TV weather presenters use graphics to show changing conditions.



37. Maze Runner Mini-Game: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students build a simple maze game where arrow keys move a sprite and a wall-touching rule ends the attempt.

What Students Do: Students use arrow-key blocks to move a sprite through a maze, test the “if touching wall” rule, and try again to improve after losing.

Learning Outcomes (Students will be able to):

- Use arrow-key blocks to move a sprite through a maze.
- Recognise an “if touching wall” rule as a simple instruction.
- Try again and improve after losing.

Concepts & Skills Covered: keyboard input, conditional rules, persistence/iteration

Real-World Relevance: This introduces the same collision-detection idea used in every maze and platform video game.



38. Feed the Virtual Pet: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students use if-then logic to build a simple virtual pet game whose pet gets hungry over time unless it is fed.

What Students Do: Students use “if-then” blocks to feed a hungry pet sprite, watch a simple hunger value change as the game runs, then describe what happens if the pet isn’t fed.

Learning Outcomes (Students will be able to):

- Use “if-then” blocks to feed a hungry pet sprite.
- Track a simple hunger value as the game runs.
- Describe what happens to the pet if it isn’t fed.

Concepts & Skills Covered: conditional logic, variables (informal), care/responsibility

Real-World Relevance: Virtual pet games use this same if-then, value-tracking idea, and it is a gentle first introduction to variables.



39. Mix-the-Colours Game: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students drag and combine colour blocks to predict, then test, what colour two colours make when mixed together.

What Students Do: Students drag colour blocks to combine two colours, predict the result before testing, and connect the activity to the earlier Rainbow Colour Mixer hardware build.

Learning Outcomes (Students will be able to):

- Drag colour blocks to combine two colours.
- Predict what colour will result before testing it.
- Connect the app to the earlier Rainbow Colour Mixer activity.

Concepts & Skills Covered: prediction, colour mixing, hardware-software connection

Real-World Relevance: Predicting-then-testing is the same basic scientific method used in real experiments.



40. Dress-Up the Robot: Programming Logic

Materials Required: Scratch (block coding)

Purpose & Objective: Students design an original robot character and switch its costume using a click event.

What Students Do: Students switch between costumes using a click event to “dress up” their robot, design an original character, then describe its features to the class.

Learning Outcomes (Students will be able to):

- Switch between costumes using a click event.
- Design an original robot character.
- Describe their character’s features to the class.

Concepts & Skills Covered: event blocks, costumes, creative design

Real-World Relevance: Character customisation like this is a core feature of most children’s educational software and games.

AGE BAND 02

JUNIOR BUILDERS

Ages 8–10

First real breadboard wiring and the bridge from picture-blocks into text-based Arduino code.



30 HARDWARE-BASED ACTIVITIES • 10 APP / WEB / GAME ACTIVITIES

6. Age Group 2 (Ages 8-10): Junior Builders

Students move from pure picture-blocks toward real circuits and their first lines of text-based logic. Most builds use the Arduino Uno on a breadboard, written first in mBlock5's hybrid block/Arduino-C view and then, from the second term onward, directly in the Arduino IDE using teacher-provided starter sketches that students complete and modify. Every project answers a concrete "who would use this, and why?" question, and students keep a one-page build log (sketch, parts list, result) for each activity, the first formal record supporting the Section 9 assessment rubric.

Primary Coding Tools: mBlock5 (block-to-code bridge), Arduino IDE (guided/starter sketches), Scratch & MIT App Inventor (software strand)

Safety Notes: Breadboard-only wiring (no soldering). Gas/flame sensor demonstrations are teacher-supervised with no live flame near students. ESD wrist straps introduced as a habit.

Part A: 30 Hardware-Based Activities (Ages 8-10)

Each session is 60 minutes and ends with the student demonstrating the finished build and explaining its real-world purpose in their own words.



1. Smart Night Light: Sensors & Environment

Materials Required: LDR sensor, LED, breadboard, Arduino Uno

Purpose & Objective: Students write their first real Arduino program that reads a light sensor and decides, using an if-statement, when to turn a lamp on.

What Students Do: Students wire the LDR and LED on a breadboard, write a short program that reads the analog sensor value, set their own brightness threshold, and test it by covering and uncovering the sensor.

Learning Outcomes (Students will be able to):

- Write a simple program that reads an analog light sensor.
- Set a brightness threshold using an if-statement.
- Explain one real-world use of automatic lighting.

Concepts & Skills Covered: analog input, if-statements, thresholds

Real-World Relevance: Automatic night lights and streetlights both use this same read-and-decide logic.



2. Touchless Hand Sanitiser Dispenser: Robotics & Actuation

Materials Required: ultrasonic sensor, SG90 servo, Arduino Uno

Purpose & Objective: Students combine a distance sensor and a servo motor in one program to build a working touchless dispenser.

What Students Do: Students wire the ultrasonic sensor and servo, write code that swings the servo when a hand is detected within a set distance, then test and adjust the triggering distance for reliability.

Learning Outcomes (Students will be able to):

- Combine a distance sensor and a servo motor in one sketch.
- Calculate a safe triggering distance through testing.
- Relate the device to hygiene technology seen in public places.

Concepts & Skills Covered: ultrasonic sensing, servo actuation, calibration

Real-World Relevance: Touchless dispensers like this became a standard hygiene fixture in public buildings worldwide.



3. Pet Door Motion Alarm: Sensors & Environment

Materials Required: PIR motion sensor, buzzer, LED, Arduino Uno

Purpose & Objective: Students program a digital input to trigger an alarm, learning to distinguish motion from no-motion states in code.

What Students Do: Students wire the PIR sensor to a digital pin, write code that sounds the buzzer and lights the LED on motion, then test it by walking past at different times and evaluate where it would help at home.

Learning Outcomes (Students will be able to):

- Program a digital input to trigger an alarm.
- Distinguish motion vs. no-motion states in code.
- Evaluate where this alarm would be useful at home.

Concepts & Skills Covered: digital input, PIR motion sensing, state detection

Real-World Relevance: The same motion-alarm logic protects pet doors, garages, and entryways in real homes.



4. Mini Weather Station Display: Sensors & Environment

Materials Required: DHT22 sensor, OLED display (I2C), Arduino Uno

Purpose & Objective: Students read live temperature and humidity data and display it clearly on a small screen using a display library.

What Students Do: Students wire the DHT22 and OLED display, write code that reads both values and prints them to the screen, then compare their own readings to a real weather report.

Learning Outcomes (Students will be able to):

- Read temperature and humidity data in code.
- Use a library to control an I2C OLED display.
- Compare their own readings to a real weather report.

Concepts & Skills Covered: sensor libraries, I2C displays, data display

Real-World Relevance: This is a miniature version of the home weather stations sold in electronics stores.



5. Plant Watering Reminder: Sensors & Environment

Materials Required: soil moisture sensor, buzzer, LED, Arduino Uno

Purpose & Objective: Students convert a raw analog moisture reading into a clear wet/dry decision using an if/else statement.

What Students Do: Students wire the soil sensor, write an if/else threshold in code that lights an LED or sounds a buzzer when soil is dry, test it in wet and dry soil, then design a watering schedule for a real plant.

Learning Outcomes (Students will be able to):

- Convert an analog moisture value into a wet/dry decision.
- Write a clear if/else threshold in code.
- Design a watering schedule for a real plant.

Concepts & Skills Covered: analog thresholds, if/else logic, plant care

Real-World Relevance: Commercial smart-garden systems use the same wet/dry threshold logic at a larger scale.



6. Line-Following Robot Basics: Robotics & Actuation

Materials Required: 2x IR sensor modules, DC motors, L298N driver, Arduino Uno

Purpose & Objective: Students wire two motors to a motor driver and write logic that lets a small robot follow a line using two IR sensors.

What Students Do: Students assemble a basic two-motor chassis, wire it to the L298N driver, write code that reads both IR sensors and steers accordingly, then test and troubleshoot incorrect turns.

Learning Outcomes (Students will be able to):

- Wire two motors to a motor driver correctly.
- Write logic that follows a line using two IR sensors.
- Troubleshoot incorrect turns by adjusting their own code.

Concepts & Skills Covered: motor drivers, IR line sensing, basic decision logic

Real-World Relevance: Line-following is the foundational skill behind real warehouse and factory delivery robots.



7. Smart Fan Controller: Sensors & Environment

Materials Required: DHT22 sensor, DC motor, L298N driver, Arduino Uno

Purpose & Objective: Students trigger a motor to run automatically based on a temperature threshold, exploring how automation can save energy.

What Students Do: Students wire the DHT22 and a motor through the L298N driver, write code that starts the fan above a chosen temperature, then test the system at different simulated temperatures.

Learning Outcomes (Students will be able to):

- Trigger a motor to run based on a temperature threshold.
- Explain how the system could save energy at home.
- Test the system at different simulated temperatures.

Concepts & Skills Covered: temperature thresholds, motor control, energy awareness

Real-World Relevance: Smart thermostats and fan controllers use this exact automatic-trigger idea to cut energy use.



8. Parking Sensor Model: Sensors & Environment

Materials Required: ultrasonic sensor, buzzer, Arduino Uno

Purpose & Objective: Students map distance readings to a changing beep frequency, recreating how real car parking sensors work.

What Students Do: Students wire the ultrasonic sensor and buzzer, write code that maps distance to beep speed, then calibrate the system for a toy or model car and explain the purpose of real parking sensors.

Learning Outcomes (Students will be able to):

- Map distance values to beep frequency in code.
- Explain the purpose of real car parking sensors.
- Calibrate the system for a toy or model car.

Concepts & Skills Covered: distance mapping, buzzer tone control, calibration

Real-World Relevance: This is the exact sensing principle used in the reversing sensors fitted to modern cars.



9. Fire Alarm Demo Kit: Power & Safety

Materials Required: flame sensor, buzzer, LED, Arduino Uno

Purpose & Objective: Students detect a flame/IR signal in code and trigger an alarm, learning why early fire detection saves lives.

What Students Do: Under teacher supervision, students wire the flame sensor and write code that sounds an alarm on detection, then observe a teacher-led flame demonstration and record their results.

Learning Outcomes (Students will be able to):

- Detect a flame/IR signal and trigger an alarm in code.
- Explain why early fire detection can save lives.
- Observe a teacher-led demonstration and record results.

Concepts & Skills Covered: flame sensing, alarms, fire safety

Real-World Relevance: Smoke and flame detectors using this same principle are required safety equipment in real buildings.



10. Rain Alert Window Closer: Sensors & Environment

Materials Required: rain sensor, SG90 servo, Arduino Uno

Purpose & Objective: Students program a servo to move only when rain is detected, modelling how a room could protect itself from rain.

What Students Do: Students wire the rain sensor and servo, write code that closes a model window when rain is detected, then adjust the sensor's sensitivity through testing.

Learning Outcomes (Students will be able to):

- Program a servo to move only when rain is detected.
- Explain how the model protects a real room from rain.
- Adjust the sensor's sensitivity through code.

Concepts & Skills Covered: rain sensing, servo actuation, sensitivity tuning

Real-World Relevance: Some real smart-home systems automatically close windows or skylights when rain is detected.



11. Sound-Activated Light Switch: Sensors & Environment

Materials Required: sound sensor (KY-037), relay module, LED, Arduino Uno

Purpose & Objective: Students use a relay to safely switch a higher-power load using a sound sensor's signal.

What Students Do: Students wire the sound sensor to control a relay-switched LED, set a reliable sound-level threshold through testing, then discuss a real situation where clap-activated lights help.

Learning Outcomes (Students will be able to):

- Use a relay to safely switch a load from a microcontroller signal.
- Set a sound-level threshold that works reliably.
- Describe a real situation where clap-activated lights help.

Concepts & Skills Covered: relays, sound thresholds, safe switching

Real-World Relevance: Relays are the standard safe way microcontrollers control higher-power household devices.



12. Reaction Timer Game: Programming Logic

Materials Required: push button, buzzer, Arduino Uno

Purpose & Objective: Students use the `millis()` function to measure elapsed time and build a fair reaction-time game.

What Students Do: Students write code using `millis()` to time how fast a player presses a button after a cue, compare scores fairly between classmates, then find and fix a timing bug in their own code.

Learning Outcomes (Students will be able to):

- Use the `millis()` function to measure elapsed time.
- Compare reaction scores between classmates fairly.
- Find and fix a timing error in their own code.

Concepts & Skills Covered: `millis()` timing, debugging, fair measurement

Real-World Relevance: This same `millis()`-based timing technique underlies real sports-timing and reflex-testing devices.



13. Mini Traffic Light Controller: Programming Logic

Materials Required: 3x LED (red/yellow/green), Arduino Uno

Purpose & Objective: Students sequence a loop with correct timing delays to control a three-light traffic sequence, then extend it with a pedestrian phase.

What Students Do: Students write a looped program with `delay()` timings for red/yellow/green, test the sequence, then modify the code to add a pedestrian-crossing phase.

Learning Outcomes (Students will be able to):

- Sequence a loop with correct timing delays.
- Explain real traffic-light timing rules.
- Modify the code to add a pedestrian-crossing phase.

Concepts & Skills Covered: loops, `delay()` timing, code modification

Real-World Relevance: Real traffic signal controllers use the same timed-sequence logic, with far more safety redundancy.



14. Automatic Pet Feeder: Robotics & Actuation

Materials Required: SG90 servo, push button, Arduino Uno

Purpose & Objective: Students program a servo to dispense a measured, repeatable portion of food.

What Students Do: Students wire the servo-based dispenser, write code that releases a measured portion on a button press, test the repeatability of the dispensing action, then relate it to real automatic feeders.

Learning Outcomes (Students will be able to):

- Program a servo to dispense a measured portion.
- Test the repeatability of the dispensing action.
- Relate the build to real automatic pet feeders.

Concepts & Skills Covered: servo calibration, repeatability, portion control

Real-World Relevance: Commercial automatic pet feeders use the same measured-dispense mechanism at consumer scale.



15. Light Theremin: Sensors & Environment

Materials Required: LDR sensor, buzzer, Arduino Uno

Purpose & Objective: Students map a light-sensor reading to a sound frequency using tone(), exploring the link between analog input and sound output.

What Students Do: Students wire the LDR and buzzer, write code using tone() to map light level to pitch, then experiment with hand position over the sensor to change the pitch and perform for the class.

Learning Outcomes (Students will be able to):

- Map a light-sensor reading to a sound frequency using tone().
- Experiment with hand position to change the pitch.
- Describe the link between analog input and sound output.

Concepts & Skills Covered: analog-to-sound mapping, tone() function, experimentation

Real-World Relevance: This introduces the same input-to-audio mapping used by real electronic musical instruments.



16. Obstacle-Avoiding Toy Car: Robotics & Actuation

Materials Required: ultrasonic sensor, DC motors, L298N driver, Arduino Uno

Purpose & Objective: Students write a decision algorithm that steers a robot away from obstacles it detects.

What Students Do: Students assemble a basic two-motor chassis, write an “if obstacle, turn” algorithm using the ultrasonic sensor, then test and refine the car’s turning behaviour.

Learning Outcomes (Students will be able to):

- Write a decision algorithm (“if obstacle, turn”) for movement.
- Assemble a basic two-motor chassis.
- Test and refine the car’s turning behaviour.

Concepts & Skills Covered: decision algorithms, obstacle detection, chassis assembly

Real-World Relevance: Obstacle avoidance is a core safety behaviour in real autonomous vehicles and warehouse robots.



17. Smart Doorbell with Display: Electronics & Circuits

Materials Required: push button, buzzer, OLED display (I2C), Arduino Uno

Purpose & Objective: Students trigger both a buzzer and an on-screen greeting message from a single button press.

What Students Do: Students wire the button, buzzer, and OLED display, write code that shows a friendly message and sounds the buzzer together, then explain the role of a doorbell in home communication.

Learning Outcomes (Students will be able to):

- Trigger a buzzer and an on-screen message from one button press.
- Design a friendly visitor-greeting message.
- Explain the role of a doorbell in home communication.

Concepts & Skills Covered: multi-output triggering, OLED text display, UX design

Real-World Relevance: Modern smart doorbells combine a chime and an on-screen message in exactly this way.



18. Digital Thermometer Display: Sensors & Environment

Materials Required: DHT22 sensor, OLED display (I2C), Arduino Uno

Purpose & Objective: Students format and print a numeric sensor reading clearly on a digital display.

What Students Do: Students wire the DHT22 and OLED, write code that formats and prints the temperature reading, then compare digital and analog thermometers and adjust the layout for readability.

Learning Outcomes (Students will be able to):

- Format and print a numeric sensor reading on a display.
- Compare digital and analog thermometers.
- Adjust the display layout for better readability.

Concepts & Skills Covered: data formatting, display layout, comparison

Real-World Relevance: Digital thermometers used in homes and clinics rely on this same read-format-display pipeline.



19. Step Counter Pedometer: Sensors & Environment

Materials Required: MPU6050 sensor, OLED display (I2C), Arduino Uno

Purpose & Objective: Students detect motion peaks from an IMU sensor and use them to count and display steps.

What Students Do: Students wire the MPU6050 and OLED, write code that detects a motion peak as a “step” and displays a running count, then relate the project to fitness trackers.

Learning Outcomes (Students will be able to):

- Detect motion peaks to count simple “steps.”
- Display a running step count on screen.

- Relate the project to fitness trackers they may have seen.

Concepts & Skills Covered: peak detection, running counters, wearable tech

Real-World Relevance: This is a simplified version of the step-counting algorithm inside every fitness tracker and smartphone.



20. Colour-Changing Mood Lamp: Electronics & Circuits

Materials Required: assorted LEDs, push button, Arduino Uno

Purpose & Objective: Students cycle through an array of colours using a button press, meeting the array data structure for the first time.

What Students Do: Students wire several LEDs, write code storing colours in an array and cycling through them on each button press, then design a mood lamp suited to a real bedroom.

Learning Outcomes (Students will be able to):

- Cycle through a list (array) of colours using a button press.
- Use an array data structure in code for the first time.
- Design a mood lamp suited to a real bedroom.

Concepts & Skills Covered: arrays, button-triggered cycling, product design

Real-World Relevance: Consumer mood lamps and smart bulbs use the same colour-array cycling approach.



21. Earthquake Shake Alarm: Sensors & Environment

Materials Required: MPU6050 sensor, buzzer, LED, Arduino Uno

Purpose & Objective: Students set a motion-threshold trigger in code, exploring how real earthquake sensors work at a basic level.

What Students Do: Students wire the MPU6050, buzzer, and LED, write code that triggers an alarm above a chosen shake threshold, then test the alarm with controlled, gentle shaking.

Learning Outcomes (Students will be able to):

- Set a motion-threshold trigger in code.
- Explain, at a basic level, how real earthquake sensors work.
- Test the alarm with controlled, gentle shaking.

Concepts & Skills Covered: motion thresholds, vibration sensing, safety testing

Real-World Relevance: Real seismic early-warning systems use the same threshold-triggered vibration-sensing principle.



22. Gas Leak Detector Demo: Power & Safety

Materials Required: MQ-2 gas sensor, buzzer, LED, Arduino Uno

Purpose & Objective: Students read an analog gas-level value and classify it as safe or unsafe in code, practising safe sensor handling.

What Students Do: Under teacher supervision, students wire the MQ-2 sensor, write code that classifies readings as safe/unsafe and triggers an alarm, then discuss why gas detectors matter in real homes.

Learning Outcomes (Students will be able to):

- Read an analog gas-level value and classify it as safe/unsafe.
- Explain why gas detectors matter in real homes.
- Practise safe handling of sensor demonstrations.

Concepts & Skills Covered: analog classification, gas sensing, lab safety

Real-World Relevance: Household and industrial gas detectors use the same analog-threshold classification approach.



23. Smart Curtain Opener: Robotics & Actuation

Materials Required: LDR sensor, SG90 servo, Arduino Uno

Purpose & Objective: Students link a sunrise-light condition to a servo action, building a miniature working curtain mechanism.

What Students Do: Students wire the LDR and servo, write code that opens the curtain mechanism once light exceeds a threshold, then test it and discuss the comfort and energy benefits of automation.

Learning Outcomes (Students will be able to):

- Link a sunrise-light condition to a servo action.
- Build a miniature working curtain mechanism.
- Explain the comfort and energy benefits of automated curtains.

Concepts & Skills Covered: light-triggered actuation, mechanism design, home automation

Real-World Relevance: Motorised curtains and blinds in smart homes use this same light-triggered opening logic.



24. Heart Rate Display Gadget: Sensors & Environment

Materials Required: heart rate pulse sensor, OLED display (I2C), Arduino Uno

Purpose & Objective: Students process a raw pulse signal into a beats-per-minute value and display it clearly.

What Students Do: Students wire the pulse sensor and OLED, write code that calculates BPM from the raw signal and displays it, then discuss healthy resting heart-rate ranges.

Learning Outcomes (Students will be able to):

- Process a raw pulse signal into a beats-per-minute value.
 - Display BPM clearly on a screen.
-

- Discuss healthy resting heart-rate ranges.

Concepts & Skills Covered: signal processing, BPM calculation, health literacy

Real-World Relevance: This same raw-signal-to-BPM pipeline runs inside every consumer heart-rate wearable.



25. Automatic Streetlamp Model: Sensors & Environment

Materials Required: LDR sensor, relay module, Arduino Uno

Purpose & Objective: Students use a relay to switch a higher-power lamp from sensor logic, modelling dusk-to-dawn automation.

What Students Do: Students wire the LDR to control a relay-switched lamp, write the switching logic, then test the model under different light conditions and explain real streetlight automation.

Learning Outcomes (Students will be able to):

- Use a relay to switch a higher-power lamp from sensor logic.
- Explain dusk-to-dawn automation used in real streetlights.
- Test the model under different light conditions.

Concepts & Skills Covered: relay switching, light thresholds, municipal automation

Real-World Relevance: City streetlight networks use this exact sensor-and-relay principle to switch on automatically at dusk.



26. Tilting Maze Game: Robotics & Actuation

Materials Required: MPU6050 sensor, 2x SG90 servos, Arduino Uno

Purpose & Objective: Students map tilt-sensor data to two servo angles to build a simple tilting maze platform.

What Students Do: Students wire the MPU6050 and two servos, write code mapping tilt readings to servo angles, then work with a partner to guide a ball through the maze.

Learning Outcomes (Students will be able to):

- Map tilt-sensor data to two servo angles.
- Build a simple tilting maze platform.
- Collaborate with a partner to guide a ball through the maze.

Concepts & Skills Covered: tilt sensing, dual-servo mapping, collaboration

Real-World Relevance: This tilt-to-actuator mapping is the same principle used in flight-simulator and camera-stabiliser platforms.



27. Library Quiet Alarm: Sensors & Environment

Materials Required: sound sensor (KY-037), 2x LED (green/red), Arduino Uno

Purpose & Objective: Students classify sound levels into “quiet” and “noisy” using thresholds, modelling a noise monitor for shared spaces.

What Students Do: Students wire the sound sensor and two LEDs, write code that classifies the noise level and lights the correct LED, then propose one improvement to the alarm’s design.

Learning Outcomes (Students will be able to):

- Classify sound levels into “quiet” and “noisy” using thresholds.
- Explain the purpose of noise monitors in shared spaces.
- Propose one improvement to the alarm’s design.

Concepts & Skills Covered: sound-level thresholds, classification, design critique

Real-World Relevance: Noise-level indicators like this are used in real libraries, hospitals, and open-plan offices.



28. Mini Elevator Model: Robotics & Actuation

Materials Required: SG90 servo, 2x push buttons, Arduino Uno

Purpose & Objective: Students program two buttons to move a platform between two positions, modelling real elevator call-button logic.

What Students Do: Students wire a servo-driven platform with two call buttons, write code that moves the platform to the requested position, then test the model for correct, reliable stopping.

Learning Outcomes (Students will be able to):

- Program two buttons to move a platform to two positions.
- Explain real elevator call-button logic.
- Test the model for correct, reliable floor-stopping.

Concepts & Skills Covered: positional control, button-to-state mapping, reliability testing

Real-World Relevance: Real elevators use the same call-and-move logic, scaled up with far greater safety systems.



29. Smart Trash Bin Lid: Robotics & Actuation

Materials Required: ultrasonic sensor, SG90 servo, Arduino Uno

Purpose & Objective: Students trigger a lid to open only within a set hand-distance range, discussing the hygiene benefits of touchless design.

What Students Do: Students wire the ultrasonic sensor and servo lid, write code that opens the lid only within a defined distance range, then adjust the detection range through testing.

Learning Outcomes (Students will be able to):

- Trigger lid-opening only within a set hand-distance range.
- Discuss the hygiene benefits of touchless bins.
- Adjust the detection range through testing.

Concepts & Skills Covered: range-bound triggering, hygiene design, calibration

Real-World Relevance: Touchless bins using this exact range-sensing idea are now common in kitchens, hospitals, and airports.



30. DIY Metronome: Programming Logic

Materials Required: buzzer, push button, Arduino Uno

Purpose & Objective: Students use a button to set a tempo variable in code and produce evenly-timed beats.

What Students Do: Students wire a buzzer and button, write code that lets a button press set a tempo variable, then use delay() to produce evenly-timed beats and relate it to real music-practice tools.

Learning Outcomes (Students will be able to):

- Use a button to set a tempo variable in code.
- Produce evenly-timed buzzer beats using delay().
- Relate the metronome to real music-practice tools.

Concepts & Skills Covered: variables, delay() timing, musical tempo

Real-World Relevance: Real digital metronomes used by musicians rely on the same variable-tempo timing logic.

Part B: 10 App / Web / Game Development Activities (Ages 8-10)

These sessions build foundational software-creation skills in parallel with the hardware track. Several explicitly extend or control an earlier hardware build, so students experience the complete hardware-to-software pipeline.



31. Catch-the-Bug Full Game: Programming Logic

Materials Required: Scratch

Purpose & Objective: Students expand a simple game with a scoring variable and a lives system, then playtest and tune its difficulty.

What Students Do: Students add a score variable and lives counter to their game, use broadcast/receive blocks to manage different game states, then playtest and adjust difficulty.

Learning Outcomes (Students will be able to):

- Add a scoring variable and a lives system to a game.
- Use broadcast/receive blocks to manage game states.
- Playtest and adjust the game's difficulty.

Concepts & Skills Covered: variables, broadcast/receive messaging, playtesting

Real-World Relevance: Score and lives systems like this are standard features in nearly every commercial video game.



32. Trivia Quiz App: Programming Logic

Materials Required: Scratch

Purpose & Objective: Students build branching if/else logic to check right and wrong quiz answers and track a running score.

What Students Do: Students write at least five original quiz questions, build if/else logic to check answers, and track and display a running score.

Learning Outcomes (Students will be able to):

- Build branching if/else logic for right and wrong answers.
- Track and display a running score.
- Write at least five original quiz questions.

Concepts & Skills Covered: if/else branching, score tracking, content authoring

Real-World Relevance: This branching-logic pattern powers real online quizzes and assessment software.



33. Physical Computing Pong: Programming Logic

Materials Required: Scratch + Arduino Uno (mBlock extension), push button

Purpose & Objective: Students connect a physical button as a real controller input for an on-screen paddle game.

What Students Do: Students wire a physical button, use the mBlock extension to synchronise the button's signal with an on-screen paddle, then explain how a physical device and a digital app connect.

Learning Outcomes (Students will be able to):

- Connect a physical button as a real game-controller input.
- Synchronise sensor data with an on-screen paddle.
- Explain how a physical device and a digital app can connect.

Concepts & Skills Covered: hardware-software bridging, serial input, game controls

Real-World Relevance: This is the same physical-to-digital link used by real game controllers and arcade cabinets.



34. My Pet Webpage: Programming Logic

Materials Required: HTML/CSS (text editor or web-based IDE)

Purpose & Objective: Students structure and style their first real webpage about a pet, using headings, paragraphs, images, and CSS.

What Students Do: Students structure a webpage using headings, paragraphs, and images, apply basic CSS styling such as colour and font, then share the finished page locally with a classmate.

Learning Outcomes (Students will be able to):

- Structure a webpage using headings, paragraphs, and images.
- Apply basic CSS styling such as colour and font.
- Share the finished page locally with a classmate.

Concepts & Skills Covered: HTML structure, CSS styling, web publishing basics

Real-World Relevance: These same HTML/CSS building blocks underlie every website on the internet.



35. Weather Dashboard Webpage: Programming Logic

Materials Required: HTML/CSS

Purpose & Objective: Students build a simple table-based webpage that displays sample weather data clearly.

What Students Do: Students build an HTML table to display sample weather data, format it clearly for a reader, and relate the page to the earlier Mini Weather Station hardware build.

Learning Outcomes (Students will be able to):

- Build a simple table to display sample weather data.
- Format data clearly for a reader.
- Relate the page to the earlier Mini Weather Station hardware build.

Concepts & Skills Covered: HTML tables, data formatting, hardware-software connection

Real-World Relevance: Real weather websites and dashboards use the same table-based approach to present data clearly.



36. Simple Calculator App: Programming Logic

Materials Required: MIT App Inventor

Purpose & Objective: Students connect buttons to basic math operations and design a clear, usable calculator layout.

What Students Do: Students connect buttons to addition, subtraction, and other operations, design a clear app layout, then test the app for incorrect or unexpected inputs.

Learning Outcomes (Students will be able to):

- Connect buttons to basic math operations.
- Design a clear and usable app layout.
- Test the app for incorrect or unexpected inputs.

Concepts & Skills Covered: event-driven buttons, UI layout, input testing

Real-World Relevance: Every calculator app on a real phone is built from these same button-to-operation connections.



37. Endless Runner Game: Programming Logic

Materials Required: Scratch

Purpose & Objective: Students build a game with increasing-speed obstacles and a clear game-over condition.

What Students Do: Students program repeated obstacles that increase in speed over time, define a clear game-over condition, then iterate on the design after playtesting.

Learning Outcomes (Students will be able to):

- Use repeated obstacles with increasing speed.
- Manage a clear game-over condition.
- Iterate on the design after playtesting.

Concepts & Skills Covered: difficulty scaling, game-over logic, iterative design

Real-World Relevance: Increasing-difficulty endless runners are one of the most popular real mobile game genres.



38. Drawing Pad App: Programming Logic

Materials Required: Scratch (pen blocks)

Purpose & Objective: Students build a simple digital drawing app using pen blocks and a colour picker.

What Students Do: Students use pen-down/pen-up blocks to draw shapes on screen, add a colour-picker feature, then create and save a simple piece of digital artwork.

Learning Outcomes (Students will be able to):

- Use pen-down/pen-up blocks to draw shapes.
- Add a colour-picker feature to the app.
- Create and save a simple piece of digital artwork.

Concepts & Skills Covered: pen blocks, colour selection, creative tools

Real-World Relevance: This is a simplified version of the drawing tools found in real digital art software.



39. Branching Story Adventure: Programming Logic

Materials Required: Scratch

Purpose & Objective: Students use variables to track a player's choices and build a story with at least two different endings.

What Students Do: Students use variables to record a player's choices through the story, build at least two different endings based on those choices, then explain how variables let a program "remember" things.

Learning Outcomes (Students will be able to):

- Use variables to track a player's choices.
- Build at least two different story endings.
- Explain how variables let a program "remember" things.

Concepts & Skills Covered: variables, branching narrative, state/memory

Real-World Relevance: Branching-choice stories use the same variable-based memory as real interactive fiction and choose-your-path apps.



40. Light-Controlled Jump Game: Programming Logic

Materials Required: Scratch + Arduino Uno (mBlock extension), LDR sensor

Purpose & Objective: Students read a live light-sensor value into Scratch and use it to trigger an on-screen jump.

What Students Do: Students wire the LDR sensor, read its live value into a Scratch variable via the mBlock extension, use the value to trigger a sprite's jump, then connect the project to earlier Light Theremin / Smart Night Light builds.

Learning Outcomes (Students will be able to):

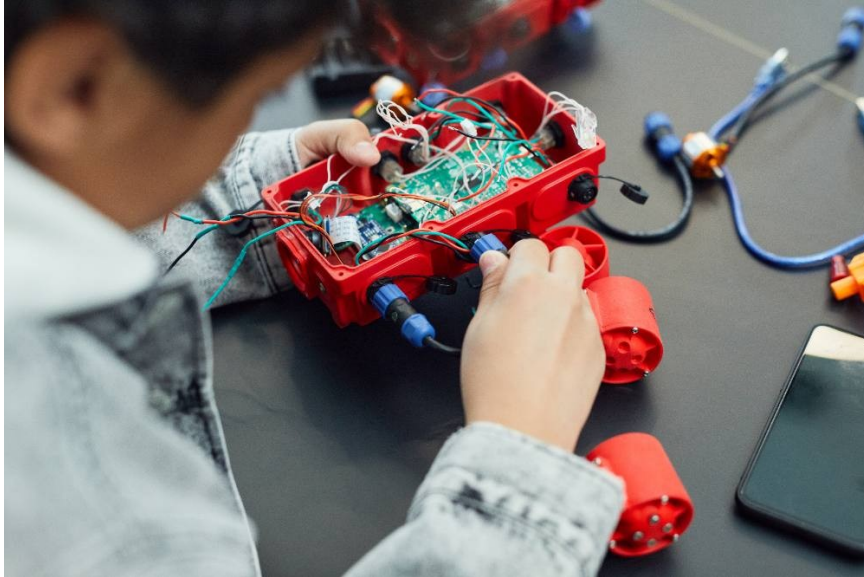
- Read a live sensor value into a Scratch variable.
- Use the sensor value to trigger a sprite's jump.
- Connect the project to the earlier Light Theremin / Smart Night Light builds.

Concepts & Skills Covered: live sensor input, hardware-software bridging, cross-activity connections

Real-World Relevance: This live-sensor-to-game-action pipeline is the same one used in real motion- and light-controlled games.

AGE BAND 03

MAKER ENGINEERS

Ages 11–15

Independent Arduino C/C++ and the program's first wireless, IoT, and multi-module builds.



30 HARDWARE-BASED ACTIVITIES • 10 APP / WEB / GAME ACTIVITIES

7. Age Group 3 (Ages 11-15): Maker Engineers

Students now write full Arduino C/C++ sketches independently in the Arduino IDE and graduate to the ESP32 for wireless and IoT work, using Thonny/MicroPython where appropriate. Activities combine two or more hardware modules, require students to justify design decisions (threshold values, wiring choices, algorithm logic) in a short written reflection, and treat debugging explicitly as a skill in its own right rather than a failure state. Several activities pair deliberately across sessions, with a hardware build's data feeding directly into a later software activity.

Primary Coding Tools: Arduino IDE (C/C++), Thonny (MicroPython on ESP32), MIT App Inventor, Python, HTML/CSS/JavaScript

Safety Notes: Multimeter use introduced for verifying connections before power-up. Flame/gas sensor activities remain supervised. ESD wrist straps required when handling ESP32/bare boards.

Part A: 30 Hardware-Based Activities (Ages 11-15)

Each session is 60 minutes and ends with the student demonstrating the finished build and explaining its real-world purpose in their own words.



1. IoT Weather Station with Web Server: IoT & Communication

Materials Required: ESP32, DHT22 sensor, OLED display, Arduino IDE

Purpose & Objective: Students configure an ESP32 to host its own web server and serve live sensor readings as a webpage, their first taste of client-server communication.

What Students Do: Students wire the DHT22 and OLED to the ESP32, write firmware that connects to Wi-Fi and serves an HTML page of live readings, then load the page from a browser on another device and explain what happened.

Learning Outcomes (Students will be able to):

- Configure ESP32 Wi-Fi and host a simple web server.
- Serve live sensor readings as a webpage.
- Explain client-server communication at a basic level.

Concepts & Skills Covered: Wi-Fi configuration, embedded web servers, client-server model

Real-World Relevance: This is the exact architecture behind every commercial “smart” weather station and IoT dashboard.



2. Bluetooth-Controlled Robot Car: IoT & Communication

Materials Required: HC-05 Bluetooth module, L298N driver, DC motors, Arduino Uno

Purpose & Objective: Students parse serial commands received over Bluetooth and map them to motor-direction logic on a driveable robot.

What Students Do: Students pair an HC-05 module with a phone or laptop, write code that parses incoming serial characters into forward/back/left/right commands, then test wireless range and command reliability.

Learning Outcomes (Students will be able to):

- Parse serial commands received over Bluetooth.
- Map received commands to motor-direction logic.
- Test wireless range and command reliability.

Concepts & Skills Covered: serial communication, command parsing, wireless control

Real-World Relevance: Bluetooth command-parsing like this underlies remote-controlled toys, drones, and telepresence robots.



3. Smart Home Light Automation: Sensors & Environment

Materials Required: PIR sensor, LDR sensor, relay module, Arduino Uno

Purpose & Objective: Students combine two sensor conditions with logical operators to control a relay-switched lamp safely.

What Students Do: Students wire the PIR and LDR sensors, write code combining both conditions with AND/OR logic before switching the relay, then evaluate the system's energy-saving potential.

Learning Outcomes (Students will be able to):

- Combine two sensor conditions using logical operators.
- Control a relay-switched lamp safely.
- Evaluate the energy-saving potential of the system.

Concepts & Skills Covered: logical operators, multi-sensor fusion, relay safety

Real-World Relevance: Combining occupancy and daylight sensors is standard practice in real commercial building-automation systems.



4. RFID Access Control System: Security & Access

Materials Required: RC522 RFID module + cards, SG90 servo, buzzer, Arduino Uno

Purpose & Objective: Students read and validate RFID card UIDs in code to trigger a lock/unlock action for authorised users only.

What Students Do: Students wire the RC522 reader, write code that reads a card's unique ID and checks it against an authorised list, trigger the servo lock for valid cards, then discuss real-world security trade-offs.

Learning Outcomes (Students will be able to):

- Read and validate RFID card UIDs in code.
- Trigger a lock/unlock servo action for authorised cards.
- Discuss real-world trade-offs of access-control security.

Concepts & Skills Covered: RFID/UID reading, whitelisting, access control logic

Real-World Relevance: This is the same core mechanism behind office keycards, hotel locks, and transit passes.



5. Obstacle-Avoiding Autonomous Robot: Robotics & Actuation

Materials Required: ultrasonic sensor, L298N driver, DC motors, Arduino Uno

Purpose & Objective: Students design a decision algorithm with multiple obstacle responses and calibrate distance thresholds experimentally.

What Students Do: Students write code offering more than one response to an obstacle (reverse, turn left, turn right), calibrate thresholds through repeated testing, then document robot performance across trials.

Learning Outcomes (Students will be able to):

- Design a decision algorithm with multiple obstacle responses.
- Calibrate sensor distance thresholds experimentally.
- Document robot performance across repeated trials.

Concepts & Skills Covered: multi-branch decision-making, experimental calibration, performance documentation

Real-World Relevance: Multi-response obstacle avoidance is a foundational behaviour in real autonomous mobile robots.



6. Wireless Sensor Network Demo: IoT & Communication

Materials Required: 2x NRF24L01 modules, DHT22 sensor, Arduino Uno (x2)

Purpose & Objective: Students configure an NRF24L01 transmitter/receiver pair over SPI to send live sensor data wirelessly between two boards.

What Students Do: Students wire and configure a transmitter and receiver pair, write code that packages and sends DHT22 data over SPI-based radio, then troubleshoot any packet loss or pairing issues.

Learning Outcomes (Students will be able to):

- Configure a transmitter/receiver pair using SPI communication.
- Transmit live sensor data wirelessly between two boards.
- Troubleshoot packet loss or pairing issues.

Concepts & Skills Covered: SPI communication, wireless pairing, packet transmission

Real-World Relevance: This transmitter/receiver pattern is the basis of real industrial and agricultural wireless sensor networks.



7. Smart Irrigation System: IoT & Communication

Materials Required: soil moisture sensor, relay module, DC motor (pump model), ESP32

Purpose & Objective: Students automate a water pump using a moisture threshold and log irrigation events with a timestamp.

What Students Do: Students wire the soil sensor and relay-driven pump model to the ESP32, write code that triggers watering below a threshold and logs each event using `millis()`, then propose water-saving improvements.

Learning Outcomes (Students will be able to):

- Automate a pump using a moisture threshold.
- Log irrigation events with a `millis()`-based timestamp.
- Propose water-saving improvements for real agriculture.

Concepts & Skills Covered: threshold automation, event logging, precision agriculture

Real-World Relevance: Precision-irrigation systems using this exact logic are used on real farms to conserve water.



8. Home Security Alarm System: Security & Access

Materials Required: PIR sensor, flame sensor, RC522 RFID module, buzzer, Arduino Uno

Purpose & Objective: Students integrate three independent sensor inputs into one alarm state machine with an RFID-based disarm sequence.

What Students Do: Students wire PIR, flame, and RFID sensors together, write a state machine that raises an alarm on any trigger and disarms only on a valid card, then evaluate false-alarm risks.

Learning Outcomes (Students will be able to):

- Integrate three sensor inputs into one alarm state machine.
- Design an RFID-based disarm sequence.
- Evaluate false-alarm risks and mitigation strategies.

Concepts & Skills Covered: state machines, multi-sensor integration, false-alarm analysis

Real-World Relevance: Multi-zone alarm panels in real homes and offices are built on exactly this state-machine principle.



9. Self-Balancing Robot Basics: Robotics & Actuation

Materials Required: MPU6050 sensor, DC motors, L298N driver, Arduino Uno

Purpose & Objective: Students read tilt-angle data from an IMU and apply a simple proportional control response to keep a robot upright.

What Students Do: Students wire the MPU6050 to a two-motor chassis, write code that reads tilt angle and applies proportional motor correction, then explain why balancing robots need fast feedback loops.

Learning Outcomes (Students will be able to):

- Read tilt-angle data from an IMU sensor.
- Apply a simple proportional control response to motors.
- Explain why balancing robots need fast sensor feedback loops.

Concepts & Skills Covered: IMU sensing, proportional control, feedback loops

Real-World Relevance: This is the same first-principles control idea behind real self-balancing scooters and Segways.



10. Gesture-Controlled Robot Arm: Robotics & Actuation

Materials Required: MPU6050 (glove), HC-05, 2-3x SG90 servos, Arduino Uno (x2)

Purpose & Objective: Students transmit glove-based gesture data wirelessly and map tilt angles to servo positions on a robot arm.

What Students Do: Students wire an MPU6050 into a glove and pair it via HC-05 with a receiving arm rig, write code mapping tilt to servo angle, then refine calibration for smoother movement.

Learning Outcomes (Students will be able to):

- Transmit gesture data wirelessly between two boards.
- Map tilt angles to servo positions.
- Refine calibration for smoother arm movement.

Concepts & Skills Covered: wearable sensing, wireless gesture transmission, servo mapping

Real-World Relevance: Gesture-controlled interfaces like this are used in real teleoperated robotics and assistive devices.



11. ESP32 IoT Air Quality Monitor: IoT & Communication

Materials Required: MQ-2 sensor, DHT22 sensor, OLED display, ESP32

Purpose & Objective: Students combine multiple sensor readings into one dashboard display and classify air quality into safety bands.

What Students Do: Students wire MQ-2 and DHT22 sensors with an OLED to the ESP32, write code that displays combined readings and classifies them into safety bands, then relate the project to real monitoring stations.

Learning Outcomes (Students will be able to):

- Combine multiple sensor readings into one dashboard display.
- Classify air-quality readings into safety bands.
- Relate the project to real environmental monitoring stations.

Concepts & Skills Covered: multi-sensor dashboards, classification bands, environmental monitoring

Real-World Relevance: City air-quality monitoring networks use the same multi-sensor, classified-band approach at scale.



12. Stepper Motor Mini Plotter: Mechanical Design

Materials Required: 2x 28BYJ-48 stepper + ULN2003 drivers, Arduino Uno

Purpose & Objective: Students control two stepper motors for coordinated X-Y movement and draw a basic shape with the mechanism.

What Students Do: Students wire two 28BYJ-48 steppers through ULN2003 drivers, write simple coordinate-based movement commands, then use the plotter to draw a basic shape.

Learning Outcomes (Students will be able to):

- Control two stepper motors for coordinated X-Y movement.
- Write simple coordinate-based movement commands.
- Draw a basic shape using the plotter mechanism.

Concepts & Skills Covered: stepper motor control, coordinate systems, coordinated motion

Real-World Relevance: This coordinated X-Y motion is the same principle used in real CNC machines, plotters, and 3D printers.



13. Smart Parking System Model: Sensors & Environment

Materials Required: 2-3x ultrasonic sensors, OLED display, Arduino Uno

Purpose & Objective: Students poll multiple distance sensors within one program and display real-time slot status.

What Students Do: Students wire two or three ultrasonic sensors and an OLED, write code that polls each sensor and displays free/occupied status, then discuss how the system could scale to a real parking lot.

Learning Outcomes (Students will be able to):

- Poll multiple distance sensors within one program.
- Display real-time slot status (free/occupied).
- Discuss scalability of the system to a real parking lot.

Concepts & Skills Covered: sensor polling, multi-input display, scalability

Real-World Relevance: Real smart-parking systems in cities and malls use the same multi-sensor polling approach.



14. Sound-Activated Room Light: Sensors & Environment

Materials Required: sound sensor, relay module, Arduino Uno

Purpose & Objective: Students calibrate a noise threshold for reliable triggering and switch a relay-controlled load from sound input.

What Students Do: Students wire the sound sensor to control a relay, calibrate a threshold that avoids false triggers, then evaluate the practicality of sound-activated automation.

Learning Outcomes (Students will be able to):

- Calibrate a noise threshold for reliable triggering.
- Switch a relay-controlled load from sound input.

- Evaluate the practicality of sound-activated automation.

Concepts & Skills Covered: threshold calibration, relay switching, practical evaluation

Real-World Relevance: Clap-on lighting systems sold commercially rely on this same calibrated sound-threshold logic.



15. Automatic Plant Care System: IoT & Communication

Materials Required: soil moisture sensor, DHT22 sensor, relay (pump), OLED display, ESP32

Purpose & Objective: Students combine three sensor inputs into one automated care routine and display system status clearly.

What Students Do: Students wire soil moisture, DHT22, and a relay-driven pump with an OLED to the ESP32, write code combining all three readings into a care routine, then design a maintenance schedule from the data.

Learning Outcomes (Students will be able to):

- Combine three sensor inputs into one automated care routine.
- Display system status clearly to a user.
- Design a maintenance schedule based on collected data.

Concepts & Skills Covered: multi-sensor automation, status display, data-driven scheduling

Real-World Relevance: Commercial smart-greenhouse systems combine exactly these sensor types into one automated routine.



16. Wireless Doorbell System: IoT & Communication

Materials Required: 2x NRF24L01 modules, push button, buzzer, Arduino Uno (x2)

Purpose & Objective: Students transmit a button-press event wirelessly between two boards with minimal delay.

What Students Do: Students wire a transmitter button unit and a receiver buzzer unit using NRF24L01 modules, write code optimised to minimise transmission delay, then compare wired vs. wireless doorbell designs.

Learning Outcomes (Students will be able to):

- Transmit a button-press event wirelessly between two boards.
- Minimise transmission delay through code optimisation.
- Compare wired vs. wireless doorbell designs.

Concepts & Skills Covered: event transmission, latency optimisation, wired vs wireless trade-offs

Real-World Relevance: Consumer wireless doorbells use this identical button-to-radio-to-chime pipeline.



17. Heart Rate Monitor with Wireless Alert: IoT & Communication

Materials Required: heart rate pulse sensor, ESP32 (Wi-Fi/Bluetooth), buzzer

Purpose & Objective: Students calculate BPM from a raw pulse signal and trigger a wireless alert when it exceeds a safe threshold.

What Students Do: Students wire the pulse sensor to an ESP32, write code that calculates BPM and sends a Wi-Fi/Bluetooth alert above a set threshold, then discuss the ethics of personal health-monitoring devices.

Learning Outcomes (Students will be able to):

- Calculate BPM from a raw pulse-sensor signal.
- Trigger a wireless alert when BPM exceeds a safe threshold.
- Discuss ethical considerations of personal health-monitoring devices.

Concepts & Skills Covered: signal processing, threshold alerting, health-data ethics

Real-World Relevance: Real wearable health devices use this same calculate-then-alert pipeline, raising the same privacy questions.



18. Line-Following Robot with Proportional Control: Robotics & Actuation

Materials Required: 2x IR sensors, L298N driver, DC motors, Arduino Uno

Purpose & Objective: Students implement a simplified proportional control loop for line-following and compare it against simple on/off control.

What Students Do: Students upgrade an existing line-follower to use proportional correction from two IR sensors, tune the control variable to reduce wobble, then compare it against on/off control performance.

Learning Outcomes (Students will be able to):

- Implement a simplified proportional control loop for line-following.
- Tune control variables to reduce wobble.
- Compare on/off control vs. proportional control performance.

Concepts & Skills Covered: proportional control, tuning, control-strategy comparison

Real-World Relevance: This is a direct stepping stone to the full PID control used in real industrial and competition robotics.



19. Smart Energy Monitor Demo: Power & Safety

Materials Required: relay module, OLED display, Arduino Uno

Purpose & Objective: Students track appliance ON-time using millis() and estimate simple energy usage from the data.

What Students Do: Students wire a relay-switched load with an OLED, write code tracking ON-time with millis(), estimate energy usage from the data, then propose one energy-saving habit.

Learning Outcomes (Students will be able to):

- Track appliance ON-time using millis().
- Estimate simple energy usage from time data.
- Propose one energy-saving habit based on the results.

Concepts & Skills Covered: time tracking, energy estimation, data-driven recommendations

Real-World Relevance: Smart plugs sold commercially track and estimate energy usage using this same time-based method.



20. Earthquake Early-Warning Demo: IoT & Communication

Materials Required: MPU6050 sensor, buzzer, ESP32 (Wi-Fi alert)

Purpose & Objective: Students detect rapid acceleration changes as a vibration event and send a Wi-Fi-based alert on detection.

What Students Do: Students wire the MPU6050 to an ESP32, write code that detects a rapid acceleration spike and sends a Wi-Fi alert message, then discuss the real-world value of even a few seconds' warning.

Learning Outcomes (Students will be able to):

- Detect rapid acceleration changes as a vibration event.
- Send a Wi-Fi-based alert message on detection.
- Discuss the real-world value of seconds of early warning.

Concepts & Skills Covered: acceleration-spike detection, wireless alerting, disaster-response value

Real-World Relevance: National seismic early-warning networks use the same detect-then-alert pattern at a much larger scale.



21. Automated Greenhouse Ventilation: Sensors & Environment

Materials Required: DHT22 sensor, SG90 servo (vent), relay (fan), Arduino Uno

Purpose & Objective: Students trigger two different actuators from one temperature threshold and design a proportionally-opening vent.

What Students Do: Students wire a DHT22 sensor with a servo vent and relay-driven fan, write code triggering both actuators together above a threshold, then evaluate the system's effect on a model greenhouse.

Learning Outcomes (Students will be able to):

- Trigger two different actuators from one temperature threshold.
- Design a vent mechanism that opens proportionally.
- Evaluate the system's effect on a model greenhouse.

Concepts & Skills Covered: multi-actuator triggering, proportional mechanisms, environmental control

Real-World Relevance: Commercial greenhouse climate-control systems coordinate vents and fans using this same principle.



22. Two-Way Wireless Data Logger: IoT & Communication

Materials Required: 2x NRF24L01 modules, DHT22 sensor, Arduino Uno (x2)

Purpose & Objective: Students design a simple two-way communication protocol and log received data with timestamps.

What Students Do: Students configure two NRF24L01-equipped boards to exchange data in both directions, write code logging each received value with a timestamp, then identify causes of transmission errors.

Learning Outcomes (Students will be able to):

- Design a simple two-way communication protocol.
- Log received data with timestamps.
- Identify causes of data transmission errors.

Concepts & Skills Covered: bidirectional communication, timestamped logging, error analysis

Real-World Relevance: Two-way telemetry links like this underpin real remote monitoring and industrial control systems.



23. RFID Library Checkout System: Security & Access

Materials Required: RC522 RFID module + cards, OLED display, Arduino Uno

Purpose & Objective: Students associate RFID UUIDs with a simple checked-out/in record in code and display checkout status.

What Students Do: Students wire the RC522 and OLED, write code that records a book's checked-out/in state against a card UUID, then discuss the privacy considerations of RFID tracking.

Learning Outcomes (Students will be able to):

- Associate RFID UUIDs with a simple checked-out/in record in code.
- Display checkout status clearly.
- Discuss privacy considerations of RFID tracking systems.

Concepts & Skills Covered: record-keeping logic, RFID association, privacy considerations

Real-World Relevance: Real library and inventory systems use this exact UUID-to-record association pattern.



24. Fire-Fighting Robot: Robotics & Actuation

Materials Required: 2-3x flame sensors, DC motors, L298N driver, buzzer, Arduino Uno

Purpose & Objective: Students use multiple flame sensors to determine direction toward a heat source and program directional movement logic.

What Students Do: Students wire two or three flame sensors around a chassis, write code comparing sensor strength to steer toward the flame, then evaluate response time and accuracy across trials.

Learning Outcomes (Students will be able to):

- Use multiple flame sensors to determine direction toward a heat source.
- Program directional movement logic.
- Evaluate robot response time and accuracy.

Concepts & Skills Covered: multi-sensor direction-finding, directional algorithms, performance evaluation

Real-World Relevance: Directional flame-seeking logic like this is studied in real disaster-response robotics research.



25. ESP32 Web-Controlled Robot: IoT & Communication

Materials Required: ESP32, L298N driver, DC motors

Purpose & Objective: Students build a webpage with control buttons hosted directly on the ESP32 and map HTTP requests to motor commands.

What Students Do: Students write ESP32 firmware serving a control webpage with directional buttons, map each button's HTTP request to a motor command, then test control latency over Wi-Fi.

Learning Outcomes (Students will be able to):

- Build a webpage with control buttons hosted on the ESP32.
- Map HTTP requests to motor commands.
- Test control latency over Wi-Fi.

Concepts & Skills Covered: embedded web servers, HTTP requests, latency testing

Real-World Relevance: This is the same browser-based control architecture used by many real consumer Wi-Fi robots and drones.



26. Smart Mailbox Notifier: IoT & Communication

Materials Required: IR or ultrasonic sensor, ESP32, buzzer

Purpose & Objective: Students detect a "mail present" event reliably and send a wireless alert when mail arrives.

What Students Do: Students wire an IR or ultrasonic sensor inside a model mailbox connected to an ESP32, write code that detects a reliable "mail present" event and sends an alert, then discuss household convenience benefits.

Learning Outcomes (Students will be able to):

- Detect a "mail present" event reliably.
- Send a Wi-Fi or local alert when mail arrives.
- Discuss convenience benefits for real households.

Concepts & Skills Covered: presence detection, wireless alerts, convenience automation

Real-World Relevance: Smart mailbox sensors sold to consumers use exactly this presence-detect-and-alert pattern.



27. Multi-Sensor Weather Dashboard: IoT & Communication

Materials Required: DHT22 sensor, rain sensor, LDR sensor, ESP32

Purpose & Objective: Students integrate three environmental sensors into a single dashboard and identify patterns across a data-collection period.

What Students Do: Students wire DHT22, rain, and LDR sensors to an ESP32, write code combining all three into one formatted dashboard, then collect data over several sessions and identify patterns.

Learning Outcomes (Students will be able to):

- Integrate three environmental sensors into a single dashboard.
- Format combined data for clarity.
- Identify patterns across a short data-collection period.

Concepts & Skills Covered: multi-sensor integration, data formatting, pattern identification

Real-World Relevance: Personal weather stations sold to hobbyists combine exactly these three sensor types.



28. Air Mouse / Gesture Pointer Prototype: Robotics & Actuation

Materials Required: MPU6050 sensor, HC-05, SG90 servo (pointer), Arduino Uno

Purpose & Objective: Students convert tilt data into pointer movement and transmit gesture data wirelessly.

What Students Do: Students wire an MPU6050-based handheld unit paired via HC-05 to a receiving servo pointer, write code converting tilt into pointer movement, then evaluate accuracy and lag.

Learning Outcomes (Students will be able to):

- Convert tilt data into pointer movement.
- Transmit gesture data wirelessly.
- Evaluate accuracy and lag of the prototype.

Concepts & Skills Covered: tilt-to-motion mapping, wireless gesture transmission, accuracy/lag evaluation

Real-World Relevance: This prototypes the same motion-sensing idea used in real air-mouse and VR-controller hardware.



29. Scheduled Automatic Pet Feeder: IoT & Communication

Materials Required: SG90/stepper, push button, ESP32 (millis-based scheduling)

Purpose & Objective: Students implement time-based scheduling without a real-time clock module and allow manual override.

What Students Do: Students wire a servo/stepper dispenser to an ESP32, write millis()-based scheduling logic that triggers feeding at set intervals with a manual-override button, then test schedule accuracy over a full session.

Learning Outcomes (Students will be able to):

- Implement time-based scheduling without a real-time clock module.
- Allow manual override via a button.
- Test schedule accuracy over a full session.

Concepts & Skills Covered: millis()-based scheduling, manual override design, accuracy testing

Real-World Relevance: This scheduling technique is a lightweight version of the timers used in real consumer smart-feeders.



30. Structural Vibration Health Monitor: Sensors & Environment

Materials Required: MPU6050 sensor, buzzer, Arduino Uno

Purpose & Objective: Students collect vibration data over time and set a justified alert threshold, relating it to real structural monitoring.

What Students Do: Students wire an MPU6050 to a model structure, write code collecting vibration data over time, set and justify an alert threshold, then relate the project to real bridge/building monitoring.

Learning Outcomes (Students will be able to):

- Collect vibration data over time.
- Set and justify an alert threshold.
- Relate the project to real bridge/building monitoring systems.

Concepts & Skills Covered: continuous data collection, threshold justification, structural health monitoring

Real-World Relevance: Real bridges and high-rise buildings are increasingly fitted with vibration sensors using this same principle.

Part B: 10 App / Web / Game Development Activities (Ages 11-15)

These sessions build foundational software-creation skills in parallel with the hardware track. Several explicitly extend or control an earlier hardware build, so students experience the complete hardware-to-software pipeline.



31. Mobile App to Control Bluetooth Robot: IoT & Communication

Materials Required: MIT App Inventor, HC-05 robot car (from earlier activity)

Purpose & Objective: Students design a custom Bluetooth-connected mobile interface to control their earlier robot car.

What Students Do: Students build a custom control screen in MIT App Inventor, send directional commands to the HC-05 robot car built earlier, then test usability with a real classmate.

Learning Outcomes (Students will be able to):

- Design a custom Bluetooth-connected control interface.
- Send directional commands from phone to robot.
- Test usability with a real user (classmate).

Concepts & Skills Covered: mobile UI design, Bluetooth app connectivity, usability testing

Real-World Relevance: This mirrors the exact app-to-hardware architecture used by real consumer Bluetooth robots and drones.



32. Personal Portfolio Website: Programming Logic

Materials Required: HTML/CSS/JavaScript

Purpose & Objective: Students structure a multi-section responsive webpage and add a simple JavaScript interaction.

What Students Do: Students structure a multi-section personal portfolio site, apply basic flexbox/grid layout techniques, then add a simple JavaScript interaction such as a button click.

Learning Outcomes (Students will be able to):

- Structure a multi-section responsive webpage.
- Apply basic CSS layout techniques (flexbox/grid basics).
- Add a simple JavaScript interaction such as a button click.

Concepts & Skills Covered: responsive layout, CSS flexbox/grid, JavaScript events

Real-World Relevance: A personal portfolio site like this is a genuine, reusable artefact for future university or job applications.



33. Python Trivia Game with Scoring: Programming Logic

Materials Required: Python (IDLE/Thonny)

Purpose & Objective: Students use functions and loops to manage a full quiz game's flow and handle invalid input gracefully.

What Students Do: Students structure the quiz using functions and loops to manage game flow, store and update a score variable, then add input validation for invalid answers.

Learning Outcomes (Students will be able to):

- Use functions and loops to manage game flow.
- Store and update a score variable.
- Handle invalid user input gracefully.

Concepts & Skills Covered: functions, loops, input validation

Real-World Relevance: Input validation like this is essential in every real production application, from forms to payment systems.



34. 2D Platformer Game: Programming Logic

Materials Required: Pygame (Python)

Purpose & Objective: Students implement sprite movement, basic gravity, and collision detection to build one complete playable level.

What Students Do: Students write Pygame code for sprite movement and simple gravity, implement collision detection between the player and platforms, then design and test one complete playable level.

Learning Outcomes (Students will be able to):

- Implement sprite movement and basic gravity.
- Detect collisions between the player and platforms.
- Design at least one complete playable level.

Concepts & Skills Covered: game physics basics, collision detection, level design

Real-World Relevance: These same physics and collision techniques are the foundation of every commercial platformer game.



35. ESP32 Sensor Dashboard Web App: IoT & Communication

Materials Required: HTML/JavaScript, ESP32 (from earlier weather station)

Purpose & Objective: Students fetch live JSON data from an ESP32 endpoint and render it with a simple live-updating chart.

What Students Do: Students write JavaScript that fetches live JSON sensor data from their earlier ESP32 weather station, render it as a simple live-updating chart, then debug a network or data-format error.

Learning Outcomes (Students will be able to):

- Fetch live JSON data from an ESP32 endpoint.
- Render sensor data with a simple live-updating chart.
- Debug a network or data-format error in the data pipeline.

Concepts & Skills Covered: JSON fetch requests, live charting, network debugging

Real-World Relevance: Fetching and charting live JSON data is the core technique behind every real IoT dashboard product.



36. To-Do List Web App: Programming Logic

Materials Required: HTML/CSS/JavaScript

Purpose & Objective: Students use the DOM to dynamically add and remove items and persist them with localStorage.

What Students Do: Students use JavaScript DOM methods to add and remove list items dynamically, store the list using localStorage so it persists, then design a clean, usable interface.

Learning Outcomes (Students will be able to):

- Use the DOM to add and remove list items dynamically.
- Store list data using localStorage.
- Design a clean, usable interface.

Concepts & Skills Covered: DOM manipulation, client-side storage, interface design

Real-World Relevance: To-do apps like this are one of the most common real starter projects for professional web developers.



37. Rule-Based Chatbot: AI & Data

Materials Required: Python

Purpose & Objective: Students implement pattern-matching response logic and structure a simple decision tree of conversation paths.

What Students Do: Students write Python code that matches user input patterns to canned responses, structure a small decision tree of conversation paths, then discuss the difference between rule-based and learning-based AI.

Learning Outcomes (Students will be able to):

- Implement pattern-matching logic for responses.
- Structure a simple decision tree of conversation paths.
- Discuss the difference between rule-based and learning-based AI.

Concepts & Skills Covered: pattern matching, decision trees, rule-based vs. learning-based AI

Real-World Relevance: Rule-based chatbots like this were the standard approach for customer-service bots before machine learning became widespread.



38. Mobile Weather App: IoT & Communication

Materials Required: MIT App Inventor, ESP32 (from earlier weather station)

Purpose & Objective: Students pull live data from their earlier ESP32 web server into a mobile app and design a clear visualisation.

What Students Do: Students connect MIT App Inventor to the earlier ESP32 weather station's data feed, design a clear visualisation for end users, then test the app on a real mobile device.

Learning Outcomes (Students will be able to):

- Pull live data from the earlier ESP32 web server into a mobile app.
- Design a clear data visualisation for end users.

- Test the app on a real mobile device.

Concepts & Skills Covered: mobile data fetching, data visualisation, device testing

Real-World Relevance: This completes the full sensor-to-phone pipeline used by every real consumer weather app.



39. Maze Pathfinding Visualiser: AI & Data

Materials Required: Python/Pygame

Purpose & Objective: Students implement a basic search algorithm to solve a maze and visualise its exploration step-by-step.

What Students Do: Students implement a breadth-first search (or similar) algorithm to solve a maze in Python/Pygame, visualise the algorithm's exploration step-by-step, then compare efficiency between two maze layouts.

Learning Outcomes (Students will be able to):

- Implement a basic search algorithm (e.g., BFS) to solve a maze.
- Visualise the algorithm's exploration step-by-step.
- Compare efficiency between two different maze layouts.

Concepts & Skills Covered: search algorithms, algorithm visualisation, efficiency comparison

Real-World Relevance: Pathfinding algorithms like this power real GPS navigation and warehouse-robot routing systems.



40. Smart Home Control App: IoT & Communication

Materials Required: MIT App Inventor or HTML, ESP32 + relay (from Smart Home activity)

Purpose & Objective: Students build an app/web interface that sends real ON/OFF commands to an ESP32-controlled relay and confirms the result.

What Students Do: Students build an interface in MIT App Inventor or HTML that sends ON/OFF commands to the earlier Smart Home relay build, implement feedback confirming the action succeeded, then evaluate the complete IoT loop.

Learning Outcomes (Students will be able to):

- Build an app/web interface that sends ON/OFF commands to an ESP32 relay.
- Implement basic user feedback to confirm status.
- Evaluate the complete IoT loop from app to physical action.

Concepts & Skills Covered: remote actuation, status feedback, end-to-end IoT evaluation

Real-World Relevance: This is the complete app-to-actuator loop used by every real commercial smart-plug and smart-switch product.

AGE BAND 04

INNOVATORS & ENGINEERS

Ages 16–18

Control theory, machine learning, and full-stack systems, built and defended like junior engineers.



30 HARDWARE-BASED ACTIVITIES • 10 APP / WEB / GAME ACTIVITIES

8. Age Group 4 (Ages 16-18): Innovators & Engineers

Students operate as junior engineers: most activities require integrating three or more hardware modules, applying control theory (PID), basic statistics or machine learning, and producing a short technical write-up or live demonstration backed by test data. The final activities in each set are open-ended capstones students scope themselves, building toward a portfolio piece suitable for a university application, national science fair, or entry to a WRO/FIRST-LEGO-League-style competition (see Section 2).

Primary Coding Tools: Arduino IDE (C/C++), MicroPython/Thonny, Python (pandas, matplotlib, scikit-learn), Node.js/Flask, HTML/CSS/JavaScript, MIT App Inventor (advanced)/Flutter, Unity/Godot/Pygame

Safety Notes: Multimeter checks required before every power-up. Battery charge/discharge labs supervised. Students follow a written test plan and log results before iterating.

Part A: 30 Hardware-Based Activities (Ages 16-18)

Each session is 60 minutes and ends with the student demonstrating the finished build and explaining its real-world purpose in their own words.



1. Full Smart Home IoT Hub: IoT & Communication

Materials Required: ESP32, multiple sensors (PIR/DHT22/LDR), relay modules, MQTT/web dashboard

Purpose & Objective: Students architect a multi-device IoT system with a central dashboard, implementing MQTT- or HTTP-based messaging between distributed nodes.

What Students Do: Students wire multiple ESP32 nodes with PIR/DHT22/LDR sensors and relays, implement a messaging protocol (MQTT or HTTP) connecting all nodes to one dashboard, then evaluate reliability and propose redundancy improvements.

Learning Outcomes (Students will be able to):

- Architect a multi-device IoT system with a central dashboard.
- Implement MQTT- or HTTP-based messaging between nodes.
- Evaluate system reliability and propose redundancy improvements.

Concepts & Skills Covered: MQTT/HTTP messaging, multi-node architecture, system reliability

Real-World Relevance: This is a scaled-down version of the exact architecture used in real commercial smart-home hub products.



2. Autonomous Obstacle-Avoidance Robot with Speed Control: Robotics & Actuation

Materials Required: ultrasonic sensor, L298N driver, DC motors, Arduino Uno/ESP32

Purpose & Objective: Students implement a proportional speed-control response so a robot slows smoothly as it nears an obstacle rather than reacting abruptly.

What Students Do: Students write code that scales motor speed proportionally to measured distance, benchmark performance across multiple runs, then optimise the code for smoother real-time decisions.

Learning Outcomes (Students will be able to):

- Implement a proportional speed-control response to obstacle distance.
- Benchmark robot performance across multiple test runs.
- Optimise code for smoother real-time decision-making.

Concepts & Skills Covered: proportional response, benchmarking, real-time optimisation

Real-World Relevance: Proportional obstacle response is a core safety behaviour in real autonomous vehicles and delivery robots.



3. Self-Balancing Two-Wheel Robot: Robotics & Actuation

Materials Required: MPU6050 sensor, DC motors, L298N driver, Arduino Uno

Purpose & Objective: Students implement a full PID control loop using IMU feedback and experimentally tune the constants for stability.

What Students Do: Students wire the MPU6050 to a two-wheel chassis, implement a PID control loop, tune P, I, and D constants experimentally, then analyse stability trade-offs between responsiveness and overshoot.

Learning Outcomes (Students will be able to):

- Implement a PID control loop using IMU feedback.
- Tune PID constants experimentally.
- Analyse stability trade-offs between responsiveness and overshoot.

Concepts & Skills Covered: PID control, empirical tuning, stability analysis

Real-World Relevance: PID-balanced two-wheel robots are the direct educational analogue of real self-balancing personal transporters.



4. Battery Management System Demo: Power & Safety

Materials Required: TP4056 charger, 18650 cells + holder, LM2596 buck converter, multimeter

Purpose & Objective: Students measure and graph a real lithium battery's charge/discharge curve and calculate voltage-regulation efficiency.

What Students Do: Students wire a TP4056 charger and 18650 cell with an LM2596 buck converter, take timed multimeter readings through a charge/discharge cycle, graph the curve, then calculate regulation efficiency under load.

Learning Outcomes (Students will be able to):

- Measure and graph a battery charge/discharge curve.
- Explain the role of protection circuitry in lithium-battery safety.
- Calculate the efficiency of voltage regulation under load.

Concepts & Skills Covered: battery charge curves, protection circuitry, regulation efficiency

Real-World Relevance: Battery-management competence like this is a genuine prerequisite skill for EV, drone, and consumer-electronics engineering roles.



5. Wireless Mesh Sensor Network: IoT & Communication

Materials Required: 3+ NRF24L01 modules, Arduino Uno (multiple), assorted sensors

Purpose & Objective: Students design a multi-node relay communication scheme and evaluate how reliability changes as node count increases.

What Students Do: Students configure three or more NRF24L01-equipped boards into a relay chain, implement basic message routing, then test and evaluate reliability as more nodes are added.

Learning Outcomes (Students will be able to):

- Design a multi-node relay communication scheme.
- Implement basic message routing between nodes.
- Evaluate network reliability as node count increases.

Concepts & Skills Covered: mesh networking, message routing, scalability testing

Real-World Relevance: Multi-hop mesh networking like this underlies real large-scale agricultural and industrial IoT deployments.



6. Secure RFID Access System with Logging: Security & Access

Materials Required: RC522 module + cards, ESP32, SG90 servo, OLED display

Purpose & Objective: Students implement timestamped access logging and a whitelist, then critically evaluate the security limits of RFID-only authentication.

What Students Do: Students wire an RC522, servo lock, and OLED to an ESP32, implement code logging every access attempt with a timestamp against a whitelist, then critically evaluate the system's security limitations.

Learning Outcomes (Students will be able to):

- Implement timestamped access logging in code.
- Design a basic authorised-user list/whitelist.
- Critically evaluate the security limitations of RFID-only authentication.

Concepts & Skills Covered: access logging, whitelisting, security critique

Real-World Relevance: Timestamped access logs are a legal requirement for many real commercial and institutional security systems.



7. Predictive Plant Health Analytics: AI & Data

Materials Required: soil moisture + DHT22 sensors, ESP32, Python (pandas/matplotlib)

Purpose & Objective: Students export logged sensor data into Python and identify trends correlating environmental data with plant stress.

What Students Do: Students log soil moisture and DHT22 data over several sessions, export it for analysis in pandas, identify correlating trends, then visualise findings in a chart suited to a non-technical audience.

Learning Outcomes (Students will be able to):

- Export logged sensor data for analysis in Python.
- Identify trends correlating soil/temperature data with plant stress.
- Visualise findings in a chart suited to a non-technical audience.

Concepts & Skills Covered: data export, correlation analysis, audience-appropriate visualisation

Real-World Relevance: This mirrors the real agritech workflow of turning field-sensor data into decisions a farmer can act on.



8. Edge Anomaly Detection for Vibration: AI & Data

Materials Required: MPU6050 sensor, ESP32 (edge processing) or Python (statistical model)

Purpose & Objective: Students establish a statistical baseline of “normal” vibration and implement a threshold- or z-score-based anomaly detector.

What Students Do: Students collect baseline vibration data from the MPU6050, calculate a statistical baseline (mean/standard deviation), implement a z-score-based detector, then evaluate false-positive/false-negative trade-offs.

Learning Outcomes (Students will be able to):

- Establish a statistical baseline of “normal” vibration from collected data.
- Implement a threshold- or z-score-based anomaly detector.
- Evaluate false-positive/false-negative trade-offs in the model.

Concepts & Skills Covered: statistical baselining, z-score anomaly detection, error-rate trade-offs

Real-World Relevance: Edge anomaly detection like this is used in real predictive-maintenance systems for machinery and infrastructure.



9. Multi-Sensor Irrigation Decision System: AI & Data

Materials Required: soil moisture, rain, DHT22 sensors, relay (pump), ESP32

Purpose & Objective: Students design and justify a weighted decision rule combining three independent sensor inputs, comparing it against simpler logic.

What Students Do: Students wire soil moisture, rain, and DHT22 sensors to an ESP32-driven pump relay, design a weighted decision rule combining all three, justify the weighting with engineering reasoning, then compare it against single-sensor logic.

Learning Outcomes (Students will be able to):

- Design a weighted decision rule combining three sensor inputs.

- Justify weighting choices with engineering reasoning.
- Compare decision-rule output against simple single-sensor logic.

Concepts & Skills Covered: weighted decision rules, engineering justification, comparative evaluation

Real-World Relevance: Weighted multi-input decision rules like this are standard practice in real precision-agriculture control systems.



10. IoT Energy Monitoring Dashboard: IoT & Communication

Materials Required: relay module, ESP32, web/cloud dashboard

Purpose & Objective: Students calculate estimated energy usage from device ON-time and rated power, and build a live dashboard of usage trends.

What Students Do: Students wire a relay-monitored appliance to an ESP32, calculate estimated energy usage from ON-time and rated power, build a live web/cloud dashboard of usage trends, then propose a data-backed cost-saving recommendation.

Learning Outcomes (Students will be able to):

- Calculate estimated energy usage from device ON-time and rated power.
- Build a live dashboard displaying usage trends.
- Propose a cost-saving recommendation backed by data.

Concepts & Skills Covered: energy estimation, live dashboards, data-backed recommendations

Real-World Relevance: This is the exact analytics loop offered by real commercial home-energy-monitoring products.



11. BLE Health Wearable Prototype: IoT & Communication

Materials Required: heart rate pulse sensor, ESP32 (BLE), small enclosure

Purpose & Objective: Students implement Bluetooth Low Energy notifications for live BPM data and design a wearable form factor.

What Students Do: Students wire the pulse sensor to an ESP32, implement BLE notifications streaming live BPM to a connected device, design a comfortable wearable enclosure, then discuss the data-privacy responsibilities of wearable health devices.

Learning Outcomes (Students will be able to):

- Implement Bluetooth Low Energy notifications for live BPM data.
- Design a wearable form factor considering comfort and power.
- Discuss the data-privacy responsibilities of wearable health devices.

Concepts & Skills Covered: BLE notifications, wearable design, health-data privacy

Real-World Relevance: This prototypes the exact BLE architecture used by every consumer fitness and health wearable on the market.



12. Robotic Arm with Inverse Kinematics Basics: Robotics & Actuation

Materials Required: MPU6050 sensor, 2-3x SG90/MG995 servos, Arduino Uno

Purpose & Objective: Students apply basic trigonometry to calculate joint angles for a target position and implement gesture-based arm control.

What Students Do: Students wire a 2-3 servo arm, calculate joint angles trigonometrically for a target position, implement gesture-based control via the MPU6050, then evaluate positional accuracy across repeated trials.

Learning Outcomes (Students will be able to):

- Apply basic trigonometry to calculate joint angles for a target position.
- Implement gesture-based control of the arm.
- Evaluate positional accuracy across repeated trials.

Concepts & Skills Covered: inverse kinematics, trigonometric calculation, positional accuracy

Real-World Relevance: Inverse kinematics is the exact mathematics used to control every real industrial and surgical robotic arm.



13. Line-Following PID Tuning Lab: Robotics & Actuation

Materials Required: IR sensor array, L298N driver, DC motors, Arduino Uno

Purpose & Objective: Students implement a full PID control loop for line-following and systematically document the effect of each constant.

What Students Do: Students wire an IR sensor array to a two-motor chassis, implement a full PID loop, systematically vary P, I, and D constants one at a time, then document the effect of each on robot behaviour.

Learning Outcomes (Students will be able to):

- Implement a full PID control loop for line-following.
- Systematically tune P, I, and D constants.
- Document the effect of each constant on robot behaviour.

Concepts & Skills Covered: full PID control, systematic tuning methodology, technical documentation

Real-World Relevance: Systematic PID tuning is a core, transferable skill used across real industrial process-control and robotics engineering.



14. Greenhouse Wireless Mesh Network: IoT & Communication

Materials Required: multiple ESP32 nodes, DHT22/soil sensors

Purpose & Objective: Students deploy multiple wireless nodes reporting to a central dashboard and implement basic node-failure detection.

What Students Do: Students deploy several ESP32 nodes with DHT22/soil sensors reporting to a central dashboard, implement a heartbeat or timeout mechanism to detect node failure, then evaluate scalability for a larger deployment.

Learning Outcomes (Students will be able to):

- Deploy multiple wireless nodes reporting to a central dashboard.
- Implement basic node-failure detection.
- Evaluate scalability for a larger agricultural deployment.

Concepts & Skills Covered: distributed node deployment, failure detection, scalability evaluation

Real-World Relevance: Node-failure detection is essential in real commercial agricultural IoT networks spanning large farms.



15. Multi-Factor Security System: Security & Access

Materials Required: PIR, RFID, flame sensor, ESP32, cloud/Wi-Fi alert

Purpose & Objective: Students combine multiple independent sensor inputs into one security state machine with a cloud-based alert mechanism, then assess it against realistic bypass scenarios.

What Students Do: Students wire PIR, RFID, and flame sensors to an ESP32, implement a combined security state machine with a cloud/Wi-Fi alert, then assess the system against realistic failure or bypass scenarios.

Learning Outcomes (Students will be able to):

- Combine multiple independent sensor inputs into one security state machine.
- Implement a cloud- or app-based alert mechanism.
- Assess the system against realistic failure or bypass scenarios.

Concepts & Skills Covered: multi-factor security logic, cloud alerting, threat assessment

Real-World Relevance: Multi-factor sensor fusion is the same principle behind real commercial and enterprise security systems.



16. CNC-Style XY Plotter: Mechanical Design

Materials Required: 2x 28BYJ-48 steppers, ULN2003 drivers, Arduino Uno

Purpose & Objective: Students implement coordinate-based motion control for two axes and parse a subset of G-code or a custom command format.

What Students Do: Students wire two steppers through ULN2003 drivers into an XY frame, implement coordinate-based motion control, parse a basic subset of G-code or a custom format, then evaluate drawing precision and propose mechanical improvements.

Learning Outcomes (Students will be able to):

- Implement coordinate-based motion control for two axes.
 - Parse a basic subset of G-code or a custom command format.
 - Evaluate drawing precision and propose mechanical improvements.
-

Concepts & Skills Covered: G-code parsing, coordinate motion control, precision evaluation

Real-World Relevance: This is a direct, functional miniature of the control systems used in real CNC routers and 3D printers.



17. Sensor Data Logging & Visualisation Pipeline: AI & Data

Materials Required: ESP32, assorted sensors, Python (pandas/matplotlib)

Purpose & Objective: Students build an automated logging pipeline from device to file and produce a clear visualisation answering a specific research question.

What Students Do: Students build an automated pipeline logging ESP32 sensor data to file, clean and process the exported data in pandas, then produce a visualisation in matplotlib answering a specific research question they define themselves.

Learning Outcomes (Students will be able to):

- Build an automated data-logging pipeline from device to file.
- Clean and process exported data in Python.
- Produce a clear visualisation answering a specific research question.

Concepts & Skills Covered: automated data pipelines, data cleaning, research-question-driven visualisation

Real-World Relevance: This end-to-end pipeline mirrors the real data-engineering workflow used in professional IoT and research settings.



18. Automation Hub with Sound-Level Triggering: IoT & Communication

Materials Required: sound sensor, relay modules, ESP32, scheduling logic

Purpose & Objective: Students combine sound-based triggers with time-based scheduling and implement priority logic when triggers conflict.

What Students Do: Students wire a sound sensor and relay modules to an ESP32, combine sound triggers with scheduling rules, implement priority logic for conflicting triggers, then evaluate the user-experience trade-offs of automated vs. manual control.

Learning Outcomes (Students will be able to):

- Combine sound-based triggers with time-based scheduling rules.
- Implement priority logic when multiple triggers conflict.
- Evaluate user-experience trade-offs of automated vs. manual control.

Concepts & Skills Covered: trigger prioritisation, scheduling logic, UX trade-off evaluation

Real-World Relevance: Conflict-resolution logic like this is essential in real multi-trigger commercial home-automation hubs.



19. Capstone: Autonomous Delivery Robot: Robotics & Actuation

Materials Required: ultrasonic, IR sensors, DC motors, L298N, RFID

Purpose & Objective: Students integrate navigation and delivery-confirmation subsystems into one robot and design a complete pickup-to-delivery routine.

What Students Do: Students combine ultrasonic/IR navigation, motor control, and RFID delivery-confirmation into one integrated robot, design and test a full pickup-to-delivery routine, then present a working demonstration with documented test results.

Learning Outcomes (Students will be able to):

- Integrate navigation and delivery-confirmation subsystems into one robot.
- Design and test a complete pickup-to-delivery routine.
- Present a working demonstration with documented test results.

Concepts & Skills Covered: subsystem integration, routine design, documented demonstration

Real-World Relevance: This capstone mirrors the real engineering challenge behind last-mile autonomous delivery robots now piloted commercially.



20. Structural Health Monitoring with Cloud Alerts: AI & Data

Materials Required: MPU6050 sensor, ESP32, cloud/Wi-Fi alert, data logging

Purpose & Objective: Students log vibration data continuously over an extended period and implement threshold-based cloud alerting.

What Students Do: Students wire an MPU6050 to an ESP32 logging continuously over an extended test, implement cloud/Wi-Fi threshold alerting, then relate findings to real bridge/building monitoring practices.

Learning Outcomes (Students will be able to):

- Log vibration data continuously over an extended test period.
- Implement threshold-based cloud alerting.
- Relate findings to real bridge/building monitoring practices.

Concepts & Skills Covered: continuous logging, cloud alerting, structural monitoring practice

Real-World Relevance: This is a direct educational analogue of the structural-health-monitoring systems deployed on real infrastructure.



21. Smart Parking Occupancy Dashboard: IoT & Communication

Materials Required: multiple ultrasonic sensors, ESP32, web dashboard

Purpose & Objective: Students aggregate multiple sensor states into one real-time dashboard and implement an occupancy-percentage calculation.

What Students Do: Students wire multiple ultrasonic sensors to an ESP32, aggregate all states into one real-time dashboard, implement an occupancy-percentage calculation, then propose a scaled deployment plan for a real facility.

Learning Outcomes (Students will be able to):

- Aggregate multiple sensor states into one real-time dashboard.
- Implement a simple occupancy-percentage calculation.
- Propose a scaled deployment plan for a real parking facility.

Concepts & Skills Covered: state aggregation, occupancy calculation, deployment planning

Real-World Relevance: This is a direct educational version of the smart-parking systems already deployed in cities and airports.



22. Wireless Robotics Telemetry System: IoT & Communication

Materials Required: 2x NRF24L01, sensors on robot, base-station dashboard

Purpose & Objective: Students transmit live robot status data to a remote base station and design a clear telemetry dashboard.

What Students Do: Students wire NRF24L01 modules to a mobile robot and a base station, transmit live sensor/status data, design a clear dashboard layout, then evaluate transmission range and reliability under different conditions.

Learning Outcomes (Students will be able to):

- Transmit live robot sensor/status data to a remote base station.
- Design a clear telemetry dashboard layout.
- Evaluate transmission range and reliability under different conditions.

Concepts & Skills Covered: telemetry transmission, dashboard design, range/reliability evaluation

Real-World Relevance: Real competition and field robots rely on exactly this kind of live telemetry link back to a human operator.



23. Buck Converter Efficiency Lab: Power & Safety

Materials Required: LM2596 buck converter, multimeter, resistive loads

Purpose & Objective: Students measure input/output voltage and current at varying loads and mathematically calculate conversion efficiency.

What Students Do: Students wire an LM2596 buck converter with resistive loads, measure input/output voltage and current at several load levels, calculate efficiency mathematically, then explain power-loss sources in switching regulators.

Learning Outcomes (Students will be able to):

- Measure input/output voltage and current at varying loads.
 - Calculate conversion efficiency mathematically.
 - Explain power-loss sources in switching regulators.
-

Concepts & Skills Covered: power measurement, efficiency calculation, switching-regulator theory

Real-World Relevance: Efficiency analysis like this is standard practice in real power-electronics and embedded-systems engineering.



24. Battery Charging Cycle Analysis: Power & Safety

Materials Required: TP4056, 18650 cells, multimeter

Purpose & Objective: Students record a full charge curve through measured intervals and analyse charge-rate and voltage-plateau behaviour.

What Students Do: Students wire a TP4056 charger to an 18650 cell, take multimeter readings at measured intervals through a full charge cycle, analyse the charge-rate and voltage-plateau behaviour, then relate findings to real battery longevity practices.

Learning Outcomes (Students will be able to):

- Record a full charge curve through measured intervals.
- Analyse charge-rate and voltage-plateau behaviour.
- Relate findings to real-world battery longevity and safety practices.

Concepts & Skills Covered: charge-curve recording, plateau analysis, battery longevity

Real-World Relevance: This same analysis technique is used by real battery engineers to validate consumer-electronics and EV battery packs.



25. ML-Based Rain Prediction Mini-Project: AI & Data

Materials Required: DHT22, LDR, rain sensor, ESP32, Python (scikit-learn)

Purpose & Objective: Students collect and label a sensor dataset for a prediction task and train a basic logistic regression model on it.

What Students Do: Students collect and label a dataset from DHT22, LDR, and rain sensors over multiple sessions, train a logistic regression model in scikit-learn, then evaluate its accuracy and discuss real-world limitations.

Learning Outcomes (Students will be able to):

- Collect and label a sensor dataset for a simple prediction task.
- Train a basic logistic regression model on the data.
- Evaluate model accuracy and discuss its real-world limitations.

Concepts & Skills Covered: dataset labelling, logistic regression, model evaluation

Real-World Relevance: This is a scaled-down version of the exact ML workflow used in real operational weather-prediction systems.



26. Full-Stack Home Automation via REST API: IoT & Communication

Materials Required: ESP32 (REST API), web client (HTML/JS)

Purpose & Objective: Students design and implement a RESTful API on the ESP32 for device control, and build a web client that consumes it.

What Students Do: Students design RESTful endpoints on the ESP32 for device control, implement them in firmware, build a web client in HTML/JS that consumes the API, then test the end-to-end system for latency and reliability.

Learning Outcomes (Students will be able to):

- Design and implement a RESTful API on the ESP32 for device control.
- Build a web client that consumes the API.
- Test the end-to-end system for latency and reliability.

Concepts & Skills Covered: REST API design, client-server integration, latency/reliability testing

Real-World Relevance: REST APIs like this are the standard integration pattern connecting virtually all real IoT devices to apps and dashboards.



27. Smart Locker Access System: Security & Access

Materials Required: RFID module, SG90 servo lock, OLED display, ESP32

Purpose & Objective: Students implement per-user access permissions and log access events, then evaluate the system against a simple threat model.

What Students Do: Students wire an RFID reader and servo lock with an OLED to an ESP32, implement per-user permissions in code, log every access event, then evaluate the system against a simple threat model.

Learning Outcomes (Students will be able to):

- Implement per-user access permissions in code.
- Log and display access events.
- Evaluate the system against a simple threat model.

Concepts & Skills Covered: permission systems, access logging, threat modelling

Real-World Relevance: Per-user permissioned lockers like this are used in real gyms, co-working spaces, and parcel-delivery systems.



28. Autonomous Fire-Response Robot: Robotics & Actuation

Materials Required: flame sensors, DC motors, L298N, relay (simulated extinguisher), buzzer

Purpose & Objective: Students design a navigation algorithm that prioritises the strongest flame signal and integrates a response action on arrival.

What Students Do: Students wire multiple flame sensors and a simulated-extinguisher relay to a mobile robot, design an algorithm prioritising the strongest signal, integrate a response action on arrival, then evaluate response time across multiple trials.

Learning Outcomes (Students will be able to):

- Design a navigation algorithm prioritising the strongest flame signal.

- Integrate a response action upon arrival at the source.
- Evaluate response time across multiple trials.

Concepts & Skills Covered: signal-strength prioritisation, response integration, trial-based evaluation

Real-World Relevance: This builds directly on the same principles used in real firefighting- and disaster-response-robotics research.



29. Biometric Health & Identity Kiosk: Security & Access

Materials Required: RFID module, heart rate pulse sensor, OLED display, ESP32

Purpose & Objective: Students combine identity verification and a basic vital-sign reading into one kiosk-style workflow and discuss the ethics of doing so.

What Students Do: Students wire an RFID reader and pulse sensor with an OLED to an ESP32, combine identity check and vital-sign reading into one workflow, design a clear kiosk-style interface, then discuss the ethical and privacy implications of combined biometric systems.

Learning Outcomes (Students will be able to):

- Combine identity verification and a basic vital-sign reading in one workflow.
- Design a clear kiosk-style user interface.
- Discuss the ethical and privacy implications of combined biometric systems.

Concepts & Skills Covered: biometric workflow integration, kiosk UI design, biometric ethics

Real-World Relevance: Combined identity-and-health kiosks like this are increasingly deployed in real corporate and healthcare settings.



30. Capstone Engineering Showcase: Robotics & Actuation

Materials Required: Student-selected combination of 4+ hardware modules from the lab inventory

Purpose & Objective: Students independently scope, design, and defend an original engineering solution integrating four or more hardware modules of their choosing.

What Students Do: Students select a real-world problem and 4+ compatible hardware modules from the lab inventory, independently design and build an integrated working prototype over an extended project period, then present and defend their design decisions to a panel using test data.

Learning Outcomes (Students will be able to):

- Independently scope and design an original engineering solution.
- Integrate multiple subsystems into one functioning prototype.
- Present and defend design decisions to a panel using test data.

Concepts & Skills Covered: independent design, subsystem integration, technical defence

Real-World Relevance: This capstone is deliberately built to produce a portfolio-ready project suitable for university applications or competition entry.

Part B: 10 App / Web / Game Development Activities (Ages 16-18)

These sessions build foundational software-creation skills in parallel with the hardware track. Several explicitly extend or control an earlier hardware build, so students experience the complete hardware-to-software pipeline.



31. Full-Stack IoT Dashboard: IoT & Communication

Materials Required: Node.js/Flask, HTML/JS/CSS, ESP32 data feed

Purpose & Objective: Students build a backend API serving live device data and a responsive frontend visualising it in real time.

What Students Do: Students build a Node.js/Flask backend API serving live ESP32 data, build a responsive frontend with real-time charts consuming it, then deploy and test the full stack end-to-end.

Learning Outcomes (Students will be able to):

- Build a backend API serving live device data.
- Build a responsive frontend with real-time charts.
- Deploy and test the full stack end-to-end.

Concepts & Skills Covered: backend API development, real-time frontend charting, full-stack deployment

Real-World Relevance: This complete backend-to-frontend pipeline is the exact architecture behind commercial IoT dashboard products.



32. ML Sensor State Classifier: AI & Data

Materials Required: Python, scikit-learn, logged sensor data

Purpose & Objective: Students prepare a labelled sensor dataset and train and evaluate a classifier, interpreting accuracy, precision, and recall.

What Students Do: Students prepare and label a sensor dataset for a classification task, train a classifier such as a decision tree or KNN in scikit-learn, then interpret accuracy, precision, and recall in context.

Learning Outcomes (Students will be able to):

- Prepare and label a sensor dataset for classification.
- Train and evaluate a classifier (e.g., decision tree/KNN).
- Interpret accuracy, precision, and recall in context.

Concepts & Skills Covered: dataset preparation, classification algorithms, evaluation metrics

Real-World Relevance: Classifying sensor states like this is a foundational technique in real industrial predictive-maintenance and IoT analytics.



33. Cross-Platform Robot Control App: IoT & Communication

Materials Required: App Inventor (advanced)/Flutter, BLE or Wi-Fi robot link

Purpose & Objective: Students design a polished multi-screen wireless control interface and iterate on it based on a real usability test.

What Students Do: Students design a multi-screen control interface in App Inventor (advanced) or Flutter, implement reliable BLE or Wi-Fi command transmission to a robot, conduct a usability test, then iterate on the design from feedback.

Learning Outcomes (Students will be able to):

- Design a polished multi-screen control interface.
- Implement reliable wireless command transmission.
- Conduct a usability test and iterate on feedback.

Concepts & Skills Covered: multi-screen UI design, wireless command reliability, usability iteration

Real-World Relevance: This professional design-test-iterate cycle mirrors how real consumer robotics and IoT apps are developed.



34. Game Development with Physical Input: Programming Logic

Materials Required: Unity/Godot or Pygame, Arduino serial input (sensor of choice)

Purpose & Objective: Students read live serial data from a microcontroller into a game engine and map it to meaningful in-game actions.

What Students Do: Students wire a sensor of choice to an Arduino sending serial data, read that live data into Unity/Godot/Pygame, map it to an in-game action, then design and playtest a complete level using the physical controller.

Learning Outcomes (Students will be able to):

- Read live serial data from a microcontroller into a game engine.
- Map physical sensor input to in-game actions.
- Design and playtest a complete game level using the physical controller.

Concepts & Skills Covered: serial-to-game-engine integration, input mapping, playtesting

Real-World Relevance: Custom physical controllers like this are exactly how real accessibility devices and novelty game controllers are built.



35. RESTful API Server for Device Management: IoT & Communication

Materials Required: Flask/Node.js, database (SQLite/JSON)

Purpose & Objective: Students design CRUD endpoints for managing multiple IoT devices and document the API for another developer.

What Students Do: Students design and implement CRUD (Create/Read/Update/Delete) endpoints in Flask/Node.js backed by a SQLite/JSON database, implement request validation and error handling, then document the API for another developer to use.

Learning Outcomes (Students will be able to):

- Design CRUD endpoints for managing multiple IoT devices.
- Implement basic request validation and error handling.
- Document the API for another developer to use.

Concepts & Skills Covered: CRUD API design, validation/error handling, technical documentation

Real-World Relevance: CRUD API design and documentation are core, everyday skills for any professional backend or IoT-platform engineer.



36. Real-Time Chat Application: Programming Logic

Materials Required: JavaScript (WebSockets), Node.js

Purpose & Objective: Students implement real-time bidirectional messaging with WebSockets and manage multiple connected clients.

What Students Do: Students implement a WebSocket server in Node.js, build a client that sends and receives messages in real time, manage multiple simultaneous connections, then handle connection/disconnection events gracefully.

Learning Outcomes (Students will be able to):

- Implement real-time bidirectional messaging using WebSockets.
- Manage multiple connected clients on a server.
- Handle connection/disconnection events gracefully.

Concepts & Skills Covered: WebSocket protocol, concurrent client management, connection lifecycle handling

Real-World Relevance: WebSocket-based real-time messaging is the same technology behind live chat, multiplayer games, and live dashboards.



37. E-Commerce Web App Prototype: Programming Logic

Materials Required: HTML/CSS/JS, backend framework (Flask/Node)

Purpose & Objective: Students implement a shopping-cart data structure and full checkout flow connected to a simple backend.

What Students Do: Students implement a shopping-cart data structure and checkout flow in HTML/CSS/JS, connect the frontend to a simple Flask/Node backend or database, then evaluate the user experience of their own prototype.

Learning Outcomes (Students will be able to):

- Implement a shopping-cart data structure and checkout flow.
- Connect a frontend to a simple backend/database.

- Evaluate the user experience of their own prototype.

Concepts & Skills Covered: cart data structures, frontend-backend integration, UX evaluation

Real-World Relevance: This prototypes the exact cart-and-checkout pattern used by every real e-commerce platform.



38. Capstone App/Dashboard for the Engineering Showcase: IoT & Communication

Materials Required: Student-selected stack, paired with Capstone Engineering Showcase hardware

Purpose & Objective: Students design a custom software interface tailored specifically to their own Capstone Engineering Showcase hardware.

What Students Do: Students design and build a software interface in their chosen stack that integrates live data or control with their capstone hardware, test the integration thoroughly, then present the complete hardware-software system together.

Learning Outcomes (Students will be able to):

- Design a custom software interface tailored to their own capstone hardware.
- Integrate live data or control between app and device.
- Present the complete hardware-software system together.

Concepts & Skills Covered: tailored interface design, hardware-software integration, capstone presentation

Real-World Relevance: Pairing custom software with custom hardware like this is exactly what a real engineering portfolio piece or start-up MVP looks like.



39. Procedural Game Level Generator: Programming Logic

Materials Required: Python/Pygame or Unity/Godot

Purpose & Objective: Students implement a procedural/randomised level-generation algorithm and balance randomness against playability.

What Students Do: Students implement a procedural level-generation algorithm in Python/Pygame or Unity/Godot, tune parameters to balance randomness with playability, then evaluate generated levels against their own design criteria.

Learning Outcomes (Students will be able to):

- Implement a procedural/randomised level-generation algorithm.
- Balance randomness with playability constraints.
- Evaluate generated levels against design criteria.

Concepts & Skills Covered: procedural generation, playability balancing, criteria-based evaluation

Real-World Relevance: Procedural generation like this is used in real commercial games to create vast worlds without hand-designing every level.



40. Cloud Database Sensor Logging App: IoT & Communication

Materials Required: Firebase/Google Sheets API, ESP32

Purpose & Objective: Students stream live sensor data into a cloud database and build a simple analytics view from stored data.

What Students Do: Students stream live ESP32 sensor data into Firebase or the Google Sheets API, build a simple analytics view (charts/summary stats) from the stored data, then evaluate cloud connectivity reliability over an extended test.

Learning Outcomes (Students will be able to):

- Stream live sensor data into a cloud database.
- Build a simple analytics view (charts/summary stats) from stored data.
- Evaluate the reliability of cloud connectivity over an extended test period.

Concepts & Skills Covered: cloud database streaming, analytics views, connectivity reliability

Real-World Relevance: Streaming sensor data to the cloud like this is the standard backbone of virtually every real commercial IoT product.

9. Assessment Framework & Rubrics

9.1 Principle

Because every activity already states its outcomes as observable “Students will be able to...” statements, assessment in this program is criterion-referenced, not norm-referenced: each student is judged against the stated outcome, not against classmates. This directly operationalises the Bloom’s Revised Taxonomy verb ladder (Anderson & Krathwohl, 2001) introduced in Section 1.

9.2 The Four-Point Activity Rubric

For every individual activity, instructors record each of the three stated Learning Outcomes on a four-point scale:

Level	Descriptor	Evidence Required
4: Achieved with Insight	Outcome met; student can also explain a limitation, extension, or real-world implication unprompted.	Working demo plus fluent, unprompted explanation.
3: Achieved	Outcome met independently, matching the “Students will be able to...” statement exactly.	Working demo plus correct explanation when asked.
2: Developing	Outcome partially met; artefact works with adult/peer support, or explanation is incomplete.	Demo works with prompting; partial explanation.
1: Not Yet	Outcome not yet met; artefact non-functional, or student cannot explain its purpose.	No working demo, or no explanation offered.

Minimum evidence per outcome, by age band: **Ages 5–7**: a working demo plus a one-line verbal explanation. **Ages 8–10**: a working demo, a verbal explanation, and a one-page build-log entry. **Ages 11–15**: a working demo, a verbal/written explanation, a build log, and a short written design-justification for at least one threshold or wiring choice. **Ages 16–18**: a working demo, a written test plan and results log, and, for capstone activities, a panel defence.

9.3 Aggregating to Program-Level Learning Outcomes

Individual activity outcomes roll up to the ten Program-Level Learning Outcomes (Section 3) once per term: an instructor reviews the rubric scores for every activity that targeted a given outcome (identifiable from the activity’s Domain and Concepts & Skills Covered fields) and records a term-level judgement of Not Yet / Developing / Achieved / Achieved with Insight for that outcome. This gives the school a defensible, aggregable record suitable for a school report, an accreditation portfolio review, or a Maarif Foundation curriculum audit, without requiring a separate testing regime layered on top of the activities themselves.

9.4 Capstone & Portfolio Assessment (Ages 16–18)

The Age Group 4 Capstone Engineering Showcase and its paired Capstone App/Dashboard activity are assessed against an expanded panel rubric covering five dimensions: problem definition and scoping, technical execution and integration, use of test data to justify design decisions, quality of the live demonstration, and quality of the defence under panel questioning. This mirrors the judging

structure used in WRO and FIRST LEGO League competitions (Section 2), so a strong capstone project is, by design, competition- and portfolio-ready without additional rework.

9.5 Recording & Reporting

A simple per-student, per-activity tracker (activity number × 4-point score per outcome) is sufficient to run this system; no specialised software is required, though the same structure maps cleanly onto any school information system that supports custom rubric-based gradebooks.

10. Implementation, Scheduling & Resourcing Guidance

10.1 Scheduling

With 30 hardware and 10 software activities per age band, each band has 40 weekly one-hour sessions, enough for a full academic year (for example, two sessions per week across roughly 20 weeks, or one session per week across 40 weeks) without repeating a single activity. Schools running a shorter robotics/AI period per week can extend the same 160-activity bank across two academic years without exhausting the content.

10.2 Materials Planning

Every activity's Materials Required field lists components by their exact inventory names (Section 4.2), so lab technicians can pull a kit per session without reinterpreting the activity. For sessions using paired boards (for example, NRF24L01 transmitter/receiver pairs, or a Bluetooth robot plus its controller board), plan for two students or two stations to share a working pair, matching the original procurement quantities.

10.3 Progression & Differentiation

Several real-world problem families intentionally reappear across age bands at increasing depth (Section 4.3). Instructors of mixed-readiness classes can substitute an activity from the band above or below to stretch or scaffold an individual student while keeping the same underlying hardware station, which keeps classroom management simple even with a wide readiness spread.

10.4 Software–Hardware Bridges

Where a Part B activity references an earlier Part A build (for example, “Mobile App to Control Bluetooth Robot” referencing the “Bluetooth-Controlled Robot Car”), schedule the hardware activity first within the same term so the physical artefact is ready for the software session to control or visualise. These cross-references are flagged directly in the relevant activity's Materials Required and Concepts & Skills Covered fields throughout Sections 5–8.

10.5 Staffing & Facilitator Preparation

Ages 5–10 activities can be delivered by a generalist STEM facilitator following the stated procedure and safety notes. Ages 11–15 benefit from an instructor comfortable reading and lightly debugging Arduino C/C++. Ages 16–18 activities (particularly the PID, machine-learning, and full-stack strands) are best delivered or co-delivered by an instructor with working knowledge of the relevant tool (Python/scikit-learn, Node.js/Flask, or a game engine); a one-time department training session per new strand is sufficient given the starter-sketch and template-driven design of the activities.

10.6 Facilities & Safety Compliance

Safety notes are stated per age band (see each age band's introduction in Sections 5–8) and escalate deliberately: from facilitator-wired circuits at Ages 5–7, through breadboard-only independent wiring at Ages 8–10, to independent multimeter-verified practice and supervised battery labs by Ages 16–18. Schools should confirm these notes align with local occupational-health-and-safety requirements for minors working with low-voltage electronics before program launch.

10.7 Competitive & Showcase Pathways

Schools wishing to extend the program beyond the classroom can field teams into WRO (its Elementary/Junior/Senior categories map cleanly onto Age Groups 2–4) or FIRST LEGO League (its Explore/Challenge divisions map onto Age Groups 1–3), using the Robotics & Actuation strand activities as direct preparation, and can use the Age Group 4 Capstone Engineering Showcase as an annual internal exhibition event for parents, the Maarif Foundation, and prospective students.

11. References

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- Further reading on project- and problem-based learning in STEM contexts, informing Section 1.1, is drawn from the broader peer-reviewed PBL/STEM-education literature: see, for example, recent systematic reviews of project-based learning in STEM education (2024–2025) indexed via *Scientific Reports* and other peer-reviewed venues.

Author's Note

I build systems that sit at the boundary between the physical and digital, robots that sense, machines that learn, and software that turns raw sensor data into something a person can actually use. That instinct, to take an abstract idea and make it tangible, is what shaped every activity in this curriculum. I wanted a five-year-old's first blinking LED and a seventeen-year-old's machine-learning capstone to feel like two points on the same continuous line, not two unrelated subjects.

Outside of curriculum design, my own work spans brain-computer interfaces, control systems, and applied machine learning, always with the same underlying question: how do we get a system to genuinely understand and respond to the world around it. That question is what I hope this program passes on to every student who works through it, not just how to wire a circuit or write a script, but how to look at a real problem and build something that solves it.

A selection of my projects and writing is collected at cassimhossain.github.io, and I am always glad to hear from educators, engineers, or students working on similar problems. I can be reached on LinkedIn at linkedin.com/in/qasimushtaq.

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